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Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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Chapter 1

Antiferromagnetic phase

In this chapter, we study antiferromagnetism in the extended Hubbard model, limiting ourselves to commensurate antiferromagnetism at half-filling. This phase is well known and studied on the square Hubbard lattice. Here we show that the insertion of the non-local attraction $\hat{H}_V$ leads to an enhancement of the antiferromagnetic (AF) instability, as a result of two cooperative effects. First, the direct amplification of the AF gap, and, second, the shrinking of electronic bandwidth that lead to the aforementioned Mott localisation.

This chapter is organised as follows. In Sec. 1.1, we present the commensurate antiferromagnetic (AF) phase and discuss its symmetries. In Secs. 1.2, 1.3, 1.4, we discuss the HF approximation in the AF phase, starting from the conventional Hubbard model and highlighting the modifications in the presence of the non-local attraction $\hat{H}_V$. In Sec. 1.5, we analyse the self-consistency equations and derive a set of constraints on the HFPs that describe the self-consistent solution. In Sec. 1.6, we present the result of numeric simulations.

1.1 The AF phase and its symmetries

Long-range antiferromagnetic ordering is defined by the insurgence of a staggered pattern of spin density with a given periodicity, with zero net magnetisation. In Fig. 1.1, we show three (of many) possible AF orderings, with the blue/red arrows indicating an abundance of spin-up/down electrons on-site. The left panel of Fig. 1.1 shows an alternating structure with two-sites periodicity in both directions. We denote this configuration as "commensurate". This structure is periodic by shifts of an amount $(1, 1)$ in lattice units. Therefore, in reciprocal space this phase has a non-zero

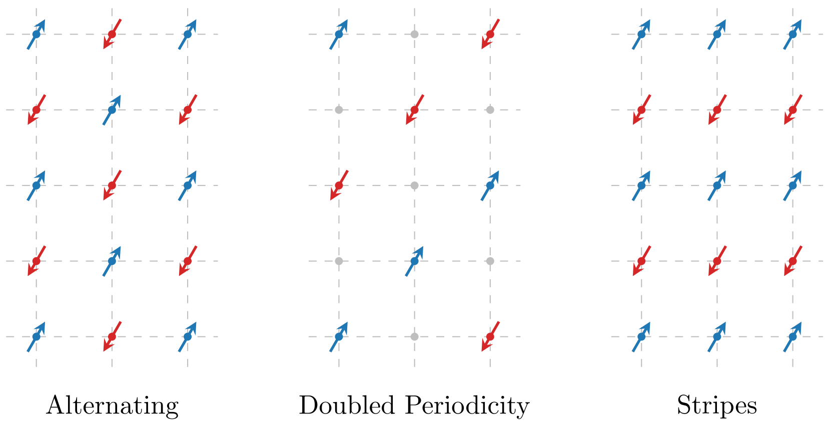


Figure 1.1 Three possible antiferromagnetic structures. Dots and arrows indicate spin abundance on a given site, while empty sites are intended as spin-balanced. Within this text we deal with the lowest-periodicity commensurate phase, the antiferromagnet on left panel.

component at wavevector

$$\boldsymbol{\pi} = (\pi, \pi)\,. \tag{1.1}$$

which we denote, from now on, the "AF vector". The central panel of Fig. 1.1 presents another staggered configuration with doubled periodicity, where sites with a given polarisation are intermediated by sites where spin populations are balanced (gray dots). The right panel shows a possible "striped" antiferromagnet, where AF ordering is established along one particular direction. In our example, in along direction y we have AF ordering with periodicity of 2 sites, while along x we have ferromagnetic ordering. Striped antiferromagnetism is studied in HM physics [28] as a possible long-range ordering energetically competing with anisotropic superconductivity. Nematic antiferromagnets, establishing long-range magnetic ordering along one particular direction, are interesting due to competition and interplay against superconducting ordering at finite doping [37]. Geometrically non-trivial magnetic ordering are at the core of SDW formation in copper oxides at finite doping [38, 39].

In this thesis, we focus on the commensurate configuration on the left panel of Fig. 1.1, which is known to be favourite at half-filling [3]. As we anticipated in Sec. **??** and summarised in Tab. **??**, the AF phase is described by the breaking of global spin rotation and translational symmetries:

$$\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)\,,$$
$$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}\,.$$

As we did for the normal phase, we also allow for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2\,.$$

We will show that the commensurate AF phase actually leaves C_4 unbroken.

Commensurate antiferromagnetism is specified by the Ansatz:

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \quad \text{and} \quad n, m \in [0,1]\,. \tag{1.2}$$

where $\mathrm{r} = (x, y) \in \mathcal{L}$. Eq. (1.2) represents the staggered distribution of spin populations shown in the left panel of Fig. 1.1. At site $\mathbf{r}$ the two spin populations average to n and differ by $2m$. Note that $\langle \hat{n}_{\mathbf{r}\sigma} \rangle \leq 1$ due to Pauli exclusion principle.

1.1.1 Symmetries of the commensurate AF Ansatz

The Ansatz of Eq. (1.2) breaks translational invariance in each spin sector, but preserves $\mathrm{U}^z(1)$ and $\mathrm{U}^c(1)$ symmetries. Eq. (1.2) is formulated equivalently as

$$\langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle = (-1)^{x+y} 2m\,. \tag{1.3}$$

Antiferromagnetism establishes when $m \neq 0$.

Proof 1.1 — The Ansatz of Eq. (1.2) is $\mathrm{U}^z(1)$-symmetric.

We use the spinors $\hat{\Psi}_{\mathbf{r}}$ defined in Eq. (**??**)

$$\hat{\Psi}_{\mathbf{r}} = \begin{bmatrix} \hat{c}_{\mathbf{r}\uparrow} \\ \hat{c}_{\mathbf{r}\downarrow} \end{bmatrix}\,,$$

and the local spin operators of Eq. (**??**),

$$\hat{S}^+_{\mathbf{r}} = \frac{\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}}{2}\,, \qquad \hat{S}^-_{\mathbf{r}} = \frac{\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}}{2}\,, \qquad \hat{S}^z_{\mathbf{r}} = \frac{\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow} + \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}}{2}\,.$$

Consider Eq. (**??**) and its conjugate,

$$\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] = -\hat{c}^\dagger_{\mathbf{r}\sigma} \quad \text{and} \quad \left[\hat{c}_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] = \hat{c}_{\mathbf{r}\sigma}\,.$$

The operators $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}^{\pm}_{\mathbf{r}}$ do not commutate. Indeed,

$$\begin{aligned}\left[\hat{S}^{+}_{\mathbf{r}}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] &= \frac{1}{2}\left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}\right]\hat{c}_{\mathbf{r}\downarrow} - \frac{1}{2}\hat{c}^{\dagger}_{\mathbf{r}\uparrow}[\hat{c}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}] \\ &= -\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\,, \\ \left[\hat{S}^{-}_{\mathbf{r}}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] &= \frac{1}{2}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}[\hat{c}_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}] - \frac{1}{2}\left[\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}\right]\hat{c}_{\mathbf{r}\uparrow} \\ &= \hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\,.\end{aligned}$$

Instead, $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}^{z}_{\mathbf{r}}$ commutate, because all on-site number operators commutate among themselves. Then, in order to have $m \neq 0$ in Eq. (1.3), SU(2) must be broken while $\mathrm{U}^z(1)$ does not. Therefore the commensurate AF phase breaks global spin rotational invariance, reducing it to the smaller symmetry of rotations around the magnetisation axis. □

Proof 1.2 — The Ansatz of Eq. (1.2) is $\mathrm{U}^{\mathrm{c}}(1)$-symmetric.

From Eq. (**??**) we know that on-site density operators are gauge-invariant separately in both spin sectors,

$$\hat{n}_{\mathbf{r}\sigma} \quad \to \quad e^{i(\eta-\eta)}\hat{n}_{\mathbf{r}\sigma} = \hat{n}_{\mathbf{r}\sigma}\,.$$

Then no selection rule applies to $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$: its EV can be non-zero without breaking $\mathrm{U}^z(1)$. □

Eq. (1.2) contains two physical observables: the average density n and the staggered magnetisation m. These two quantities obey the self-consistency equations:

$$n = \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow}\rangle + \langle \hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\,, \tag{1.4}$$

$$m = \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y}\frac{\langle \hat{n}_{\mathbf{r}\uparrow}\rangle - \langle \hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\,, \tag{1.5}$$

where the expectation values on the right-hand side depend on n and m. Since we work at fixed density, Eq. (1.4) is used to determine self-consistently the chemical potential.

Proof 1.3 — Proof of Eqs. (1.4) and (1.5).

The on-site averaged density is given by:

$$\forall \mathbf{r} \in \mathcal{L} \quad : \quad n \equiv \frac{\langle \hat{n}_{\mathbf{r}\uparrow}\rangle + \langle \hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\,, \tag{1.6}$$

while on-site magnetisation is, inverting Eq. (1.3),

$$\forall \mathbf{r} = (x,y) \in \mathcal{L} \quad : \quad m \equiv (-1)^{x+y}\frac{\langle \hat{n}_{\mathbf{r}\uparrow}\rangle - \langle \hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\,. \tag{1.7}$$

By construction, both n and m are translationally invariant quantities. Hence

$$\begin{aligned} n &= \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow}\rangle + \langle \hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\,, \\ m &= \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y}\frac{\langle \hat{n}_{\mathbf{r}\uparrow}\rangle - \langle \hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\,, \end{aligned}$$

which ends the proof. □

1.1.2 The AF Ansatz in reciprocal space

Translational symmetry is not completely broken in the AF phase; rather, is reduced. Therefore, it is useful to move to reciprocal space and transform Eq. (1.2). We define

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \,, \tag{1.8}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma} \,, \tag{1.9}$$

and their expectation values (EVs),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{n}_{\mathbf{k}\sigma} \rangle \,, \tag{1.10}$$

$$v_{\mathbf{k}\sigma} \equiv i\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \,. \tag{1.11}$$

In the definition of $v_{\mathbf{k}\sigma}$ we included a i prefactor that is useful in later calculations. Using the functions of Tab. **??**, we define the harmonics

$$u^{(\gamma)}_{\sigma} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} u_{\mathbf{k}\sigma} \,, \tag{1.12}$$

$$v^{(\gamma)}_{\sigma} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} v_{\mathbf{k}\sigma} \,. \tag{1.13}$$

As will be discussed thoroughly, these harmonics are HFPs in the AF phase and must be determined self-consistently. Eqs. (1.12) and (1.13) are their self-consistency equations.

A finite staggered magnetisation ($m \neq 0$) implies, at least for a subset of momenta $\{\mathbf{k}\}$,

$$m \neq 0 \quad \Longrightarrow \quad \exists \mathbf{k} \in \mathrm{BZ} \quad : \quad \left| \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \right| > 0 \,. \tag{1.14}$$

Removing one electron at $(\mathbf{k}, \sigma)$ and creating another at $(\mathbf{k} + \boldsymbol{\pi}, \sigma)$ produces a state with a finite overlap with the ground-state.

Proof 1.4 — If $m \neq 0$, for some $\mathbf{k}$ it must hold $\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \neq 0$.

We assume $m \neq 0$. From Eq. (1.2) we get

$$\begin{aligned}
(-1)^{\delta_{\sigma=\downarrow}} m &= \overbrace{\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} n}^{=0} + \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{\delta_{\sigma=\downarrow}} m \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} \left(n + (-1)^{x+y+\delta_{\sigma=\downarrow}} m \right) \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \,,
\end{aligned} \tag{1.15}$$

where we used $1 = \left((-1)^{x+y}\right)^2$. $\mathcal{F}$-transforming the operator and its sign factor inside the sum in last line, we get

$$\begin{aligned}
\sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} \hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma} \\
&\overset{\mathcal{F}}{=} \sum_{\mathbf{r}\in\mathcal{L}} e^{i\boldsymbol{\pi}\cdot\mathbf{r}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}} \hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\mathbf{k}'\in\mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{-i[\mathbf{k}'-(\mathbf{k}+\boldsymbol{\pi})]\cdot\mathbf{r}} \,.
\end{aligned}$$

Using Eq. (**??**), last line is reduced to

$$\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} = \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \tag{1.16}$$

$$= \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{\psi}_{\mathbf{k}\sigma}\,. \tag{1.17}$$

We insert the last line in Eq. (1.15),

$$(-1)^{\delta_{\sigma=\downarrow}} m = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle\hat{\psi}_{\mathbf{k}\sigma}\rangle\,. \tag{1.18}$$

Since $m \neq 0$, at least for some momentum $\mathbf{k}$ it must hold $\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle \neq 0$. □

The self-consistency equation for m reads, in reciprocal space,

$$m = \frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle\hat{\psi}_{\mathbf{k}\uparrow} - \hat{\psi}_{\mathbf{k}\downarrow}\rangle\,. \tag{1.19}$$

This is the transformed version of Eq. (1.5).

Proof 1.5 — Proof of Eq. (1.19): $\mathcal{F}$-transform of Eq. (1.5).

From Eq. (1.18), we have

$$m = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle\hat{\psi}_{\mathbf{k}\uparrow}\rangle \tag{1.20}$$

$$= -\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle\hat{\psi}_{\mathbf{k}\downarrow}\rangle\,. \tag{1.21}$$

Summing the two lines, we obtain Eq. (1.19). □

The operator $\hat{\psi}_{\mathbf{k}\sigma}$ satisfies the relation

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}\hat{\psi}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} = -\varphi^{(\gamma)}_{\mathbf{k}}\hat{\psi}_{\mathbf{k}\sigma}\,, \tag{1.22}$$

where we used the harmonics $\varphi^{(\gamma)}_{\mathbf{k}}$ defined in Tab. **??**.

Proof 1.6 — Proof of Eq. (1.22).

The following identity holds

$$\begin{aligned}\hat{\psi}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} &= \left(\hat{c}^{\dagger}_{\mathbf{k}+2\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right)^{\dagger}\\ &= \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma}\\ &= \hat{\psi}_{\mathbf{k}\sigma}\,.\end{aligned} \tag{1.23}$$

Moreover, being combinations of sines and cosines, the functions $\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}$ are antisymmetric under $\boldsymbol{\pi}$ shifts,

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{k}}\,. \tag{1.24}$$

Combining Eq. (1.23) and (1.24), we obtain Eq. (1.22). □

We recall now Tabs. **??** and **??**. The contractions that, through Wick's theorem, possibly give non-zero contributions are

$\hat{H}_U$	Hartree contractions,
$\hat{H}_V$ (o.s. sector)	Hartree contractions,
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions.

Operator	Sector	Channel	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$, opening of an AF gap
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalisation and resymmetrization, gap complexification

Table 1.1 | Wick's contractions for each interaction and channel in the HF framework, and their net effect in the AF phase.

Tab. 1.1 summarises the roles of the various contraction channels, which are described in next sections.

1.2 The AF phase in the conventional HM

In this section, we set $V = 0$ and describe antiferromagnetism in the conventional HM, $\hat{H}_t + \hat{H}_U$. The two-body repulsive interaction is

$$\hat{H}_U = U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} \,.$$

We know from Tab. **??** that the only contribution to the HF hamiltonian comes from Hartree channel, while Fock and Cooper channel are identically zero in the AF phase.

1.2.1 Effects of the Hartree terms: chemical potential renormalisation and magnetisation

In Eq. (**??**) we decomposed $\hat{H}_U$ through Wick's theorem. The decoupling in the Hartree channel reads

$$\hat{H}_U \overset{\text{HF}}{\simeq} U \sum_{\mathbf{r}\in\mathcal{L}} \left[\hat{n}_{\mathbf{r}\uparrow}\langle\hat{n}_{\mathbf{r}\downarrow}\rangle + \langle\hat{n}_{\mathbf{r}\uparrow}\rangle\hat{n}_{\mathbf{r}\downarrow} - \langle\hat{n}_{\mathbf{r}\uparrow}\rangle\langle\hat{n}_{\mathbf{r}\downarrow}\rangle\right] . \tag{1.25}$$

We define the MBZ as in Eq. (**??**), as the set of points in reciprocal space that satisfy the inequality $|k_x| + |k_y| \le \pi$. We sketched it in Fig. **??**. Applying the Ansatz of Eq. (1.2), we obtain the HF approximation

$$\hat{H}_U \overset{\text{HF}}{\simeq} -\sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}\left(\Delta_{\mathbf{k}\sigma}\hat{\psi}_{\mathbf{k}\sigma} + \Delta^*_{\mathbf{k}\sigma}\hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right) + nU\hat{N} - E^{(\text{AF})}_{\text{H},U} \,, \tag{1.26}$$

where we defined the gap function

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU \,, \tag{1.27}$$

ans the energy shift

$$E^{(\text{AF})}_{\text{H},U} \equiv U \sum_{\mathbf{r}\in\mathcal{L}} \langle\hat{n}_{\mathbf{r}\uparrow}\rangle\langle\hat{n}_{\mathbf{r}\downarrow}\rangle \,. \tag{1.28}$$

We observe, in Eq. (1.26), the presence of a number operator, meaning that chemical potential is shifted in the Hartree channel by

$$\mu \to \mu - nU \,. \tag{1.29}$$

Proof 1.7 — Proof of Eq. (1.26): HF approximation of $\hat{H}_U$ in Hartree channel.

Consider Eq. (1.25). Using Eqs. (1.4), (1.5) we can re-write

$$\begin{aligned}
\hat{n}_{\mathbf{r}\uparrow}\langle\hat{n}_{\mathbf{r}\downarrow}\rangle + \langle\hat{n}_{\mathbf{r}\uparrow}\rangle\hat{n}_{\mathbf{r}\downarrow} &= \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle + \langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\left(\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\right) - \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle - \langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\left(\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right)\\
&= n\left(\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\right) - m\left(\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right) .
\end{aligned}$$

Inserting this decomposition and the expression for $E^{(\mathrm{AF})}_{\mathrm{H},U}$ given in Eq. (1.28) into Eq. (1.25), we obtain

$$\hat{H}_U \overset{\mathrm{HF}}{\simeq} nU \sum_{\mathbf{r}\in\mathcal{L}} \left(\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\right) - mU \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} \left(\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right) - E^{(\mathrm{AF})}_{\mathrm{H},U} . \tag{1.30}$$

The first term gives a number operator,

$$nU \sum_{\mathbf{r}\in\mathcal{L}} \left(\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\right) = nU \times \hat{N} .$$

Using Eq. (1.16), the second term of Eq. (1.28) becomes

$$mU \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} \left(\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right) = mU \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} (-1)^{\delta_{\sigma=\downarrow}} \hat{\psi}_{\mathbf{k}\sigma} .$$

Recall the definition of the MBZ given in Eq. (**??**). The full BZ can be seen as the sum of two MBZ shifted by $\boldsymbol{\pi}$. Then, from Eq. (1.23), we get

$$mU \sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} \left(\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right) = mU \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} (-1)^{\delta_{\sigma=\downarrow}} \left(\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right) .$$

Composing everything, Eq. (1.30) becomes

$$\hat{H}_U \overset{\mathrm{HF}}{\simeq} -mU \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} (-1)^{\delta_{\sigma=\downarrow}} \left(\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right) + nU\hat{N} - E^{(\mathrm{AF})}_{\mathrm{H},U} .$$

By defining the gap function $\Delta_{\mathbf{k}\sigma}$ as in Eq. (1.27), we obtain the HF approximation of Eq. (1.26). □

The contraction energy, defined in Eq. (1.28), is given by

$$E^{(\mathrm{AF})}_{\mathrm{H},U} = L_x L_y \times U \left(n^2 - m^2\right) . \tag{1.31}$$

Compared with that for the normal phase given in Eq. (**??**), we observe the presence of an additional term dependent on m^2.

Proof 1.8 — Proof of Eq. (1.31): local Hartree contraction energy.

Inserting Eq. (1.2) in Eq. (1.28), we obtain

$$\begin{aligned} E^{(\mathrm{AF})}_{\mathrm{H},U} &\equiv U \sum_{\mathbf{r}\in\mathcal{L}} \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= U \sum_{\mathbf{r}\in\mathcal{L}} \left(n + (-1)^{x+y} m\right) \left(n - (-1)^{x+y} m\right) . \end{aligned}$$

Expanding the product above we get:

$$\begin{aligned} \left(n + (-1)^{x+y} m\right) \left(n - (-1)^{x+y} m\right) &= n^2 + \left((-1)^{x+y} - (-1)^{x+y}\right) nm - \left((-1)^{x+y}\right)^2 m^2 \\ &= n^2 - m^2 . \end{aligned}$$

The sum over $\mathcal{L}$ gives $L_x L_y$ identical terms. Hence we obtain Eq. (1.31). □

1.2.2 Antiferromagnetism in the HM

One of the aims of this projects is to identify, theoretically, what changes for each phase due to the presence of the non-local attraction $\hat{H}_V$ with respect to the conventional HM. In this section, we give a description of antiferromagnetism in the conventional HM, and later we move to the EHM.

Matrix representation. The HM hamiltonian contains the kinetic operator and the local repulsion,

$$\hat{H}_t + \hat{H}_U \,.$$

For the AF phase, the HF hamiltonian can be written as

$$\hat{H}_t + \hat{H}_U \overset{\text{HF}}{\simeq} \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + nU\hat{N} - E^{(\text{AF})}_{\text{H},U} \,, \tag{1.32}$$

where we defined the 2×2 matrix:

$$h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta^*_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} , \tag{1.33}$$

and the spinors

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} . \tag{1.34}$$

Proof 1.9 — Proof of Eq. (1.32).

Recall Eq. (**??**): the kinetic sector is represented in reciprocal space by

$$\begin{aligned} \hat{H}_t &= \sum_{\mathbf{k}\in\text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \left(\hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right) , \end{aligned}$$

where we used antiperiodicity by $\boldsymbol{\pi}$ of the bare bands $\epsilon_{\mathbf{k}}$ defined in Eq. (**??**),

$$\epsilon_{\mathbf{k}} = -\epsilon_{\mathbf{k}+\boldsymbol{\pi}} \,.$$

The kinetic operator can be written in terms of the spinors $\hat{\Psi}_{\mathbf{k}\sigma}$ defined in Eq. (1.34),

$$\hat{H}_t = \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & \\ & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \hat{\Psi}_{\mathbf{k}\sigma} \,.$$

The local repulsion $\hat{H}_U$ of Eq. (1.26) can also be written in terms of these spinors:

$$\hat{H}_U = \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \begin{bmatrix} & -\Delta^*_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & \end{bmatrix} \hat{\Psi}_{\mathbf{k}\sigma} + nU\hat{N} - E^{(\text{AF})}_{\text{H},U} \,.$$

Summing $\hat{H}_t + \hat{H}_U$ and defining $h_{\mathbf{k}\sigma}$ as in Eq. (1.33), we obtain the matricial representation of Eq. (1.32). □

Pseudospin representation. The matrix $h_{\mathbf{k}\sigma}$ can be written in term of the Pauli matrices τ^{α}

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma}\tau^z - \Delta_{\mathbf{k}\sigma}\tau^x \,.$$

Let us define the pseudospin operator $\hat{\mathbf{s}}_{\mathbf{k}\sigma}$

$$\hat{s}^{\alpha}_{\mathbf{k}\sigma} \equiv \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \tau^{\alpha} \hat{\Psi}_{\mathbf{k}\sigma} \quad \text{with} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}\sigma} \\ \hat{s}^y_{\mathbf{k}\sigma} \\ \hat{s}^z_{\mathbf{k}\sigma} \end{bmatrix} , \tag{1.35}$$

and the pseudofield $\mathbf{b}_{\mathbf{k}\sigma}$:

$$\mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix} .$$

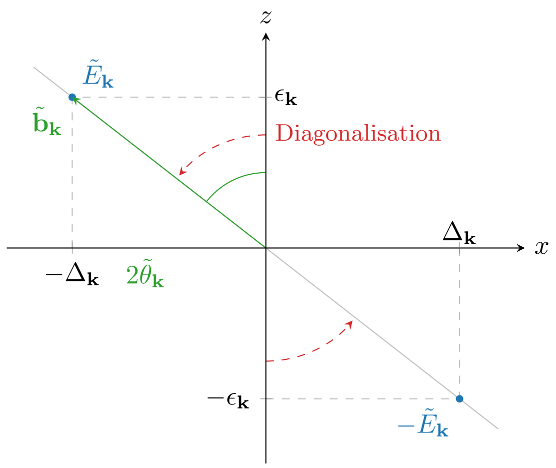


Figure 1.2 | Sketch of the diagonalisation procedure for the HM antiferromagnet. To diagonalise the HF hamiltonian, we align the z axis with the pseudo-field.

These operators satisfy a $\mathfrak{su}(2)$ algebra up to a constant C,

$$\left[\hat{s}^{\alpha}_{\mathbf{k}\sigma}, \hat{s}^{\beta}_{\mathbf{k}\sigma}\right] = C \times i \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \hat{s}^{\gamma}_{\mathbf{k}\sigma} \,. \tag{1.36}$$

The proof of Eq. (1.36) is identical to the one presented in Sec. **??**. Let us express the pseudospin operators in terms of the operators of Eqs. (1.8), (1.9):

$$\hat{s}^{x}_{\mathbf{k}\sigma} = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^{\dagger}_{\mathbf{k}\sigma} \,, \tag{1.37}$$

$$\hat{s}^{y}_{\mathbf{k}\sigma} = i\left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right) \,, \tag{1.38}$$

$$\hat{s}^{z}_{\mathbf{k}\sigma} = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \,. \tag{1.39}$$

We report these relations also in Tab. 1.2. Let also:

$$\begin{aligned} \hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \mathbb{I}_{2\times 2} \hat{\Psi}_{\mathbf{k}\sigma} \\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \,. \end{aligned} \tag{1.40}$$

Using the pseudospin representation, the hamiltonian of Eq. (1.32) can be expressed as

$$\hat{H}_t + \hat{H}_U \overset{\text{HF}}{\simeq} \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} + nU\hat{N} - E^{(\text{AF})}_{\text{H},U} \,, \tag{1.41}$$

describing a set of (pseudo)spins subject to local (pseudo)magnetic fields. The field $\mathbf{b}_{\mathbf{k}\sigma}$ is orthogonal to the y plane, and is tilted in the z plane by an angle $2\theta_{\mathbf{k}\sigma}$, where

$$\cos 2\theta_{\mathbf{k}\sigma} = \frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \,, \qquad \sin 2\theta_{\mathbf{k}\sigma} = \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \,, \qquad E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2} \,.$$

A sketch of these angles and the pseudofield is given in Fig. 1.2.

We observe that we could have redefined the pseudospin operators of Eq. (1.35) dividing them by an appropriate constant, in order to satisfy exactly the $\mathfrak{su}(2)$ algebra, and at the same time multiply the pseudo-magnetic field by the same constant. The product $\mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma}$ would be unchanged, and the diagonalisation matrix $W_{\mathbf{k}\sigma}$ would be the same. Because of this, the constant C in Eq. (1.36) is irrelevant.

Diagonalisation. In order to diagonalise the HF hamiltonian, Eq. (1.41), we perform a unitary operation on the $\hat{\Psi}_{\mathbf{k}\sigma}$ spinors which reduces the 2×2 matrix $h_{\mathbf{k}\sigma}$ of Eq. (1.33) to a diagonal form. This means to diagonalise each $h_{\mathbf{k}\sigma}$. This is obtained through a rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W^{\dagger}_{\mathbf{k}\sigma} \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix} \,. \tag{1.42}$$

The matrix $W_{\mathbf{k}\sigma}$ is unitary, and gives a counter-clockwise rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis:

$$\begin{aligned} W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos\theta_{\mathbf{k}\sigma} - i\sin\theta_{\mathbf{k}\sigma}\tau^y \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} . \end{aligned} \tag{1.43}$$

Its eigenvalues are $\pm E_{\mathbf{k}\sigma}$, with

$$E_{\mathbf{k}\sigma} = \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2} . \tag{1.44}$$

In Fig. 1.2 we sketch the pseudo-magnetic field in the xz plane, and the diagonalisation procedure. $W_{\mathbf{k}\sigma}$ is a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, and in the end aligns the original z axis with the pseudofield axis. We notice that $E_{\mathbf{k}\sigma}$ is actually spin independent: only the gap depends on σ through a sign factor, which is irrelevant in Eq. (1.44).

Applying the rotation matrix to the spinors we obtain new spinors of $\gamma^{(\pm)}$ fermions,

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix} . \tag{1.45}$$

Bringing all together, we obtain the non-interacting hamiltonian:

$$\begin{aligned} \hat{H}_t + \hat{H}_U &\overset{\text{HF}}{\simeq} \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + (\cdots) \\ &= \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} W_{\mathbf{k}\sigma}^{\dagger} W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W_{\mathbf{k}\sigma}^{\dagger} W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + (\cdots) \\ &= \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Gamma}_{\mathbf{k}\sigma}^{\dagger} d_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} + (\cdots) \\ &= \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} E_{\mathbf{k}\sigma} \left(\hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} - \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \right) + (\cdots) , \end{aligned}$$

where we intend $(\cdots) = nU\hat{N} - E_{\text{H},U}^{(\text{AF})}$. The final non-interacting hamiltonian describes a free Fermi gas of $\gamma_{\mathbf{k}\sigma}^{(\pm)}$ fermions, with energies $\pm E_{\mathbf{k}\sigma}$.

1.2.3 Features of the AF solution

This subsection is devoted to discussing briefly some features of the AF solution we found. It is organised as a series of unrelated paragraphs, each aimed at deriving precise properties or formulas we use in the algorithmic computation. Additional informations about antiferromagnetism in the conventional HM are given in App. **??**. With these observations we conclude the part about the local repulsion and expand to full EHM.

Pseudospin EVs. The expectation value of the original pseudospin operators in the various directions can be expressed in terms of the tilted pseudospins $\hat{\Gamma}_{\mathbf{k}\sigma}^{\dagger}\tau^{\alpha}\hat{\Gamma}_{\mathbf{k}\sigma}$,

$$\begin{aligned} \langle \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= -\sin 2\theta_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^{\dagger} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle , \\ \langle \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= 0 , \\ \langle \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= \cos 2\theta_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^{\dagger} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle , \end{aligned}$$

where we used the angular relations of Fig. 1.2. Moving to the tilted frame, the z component of the pseudo-spin reduces to the thermal occupation of "up" and "down" states

$$\langle \hat{\Gamma}_{\mathbf{k}\sigma}^{\dagger} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = f\left(E_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) - f\left(-E_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) . \tag{1.46}$$

Let us define the thermal factors:

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}\sigma}[\beta,\mu] \equiv f\left(-E_{\mathbf{k}\sigma};\beta,\mu\right) \pm f\left(E_{\mathbf{k}\sigma};\beta,\mu\right)\,, \tag{1.47}$$

which will often appear in computations. Being $E_{\mathbf{k}\sigma}$ spin independent, so are the thermal factors. We have:

$$\langle \hat{s}^x_{\mathbf{k}\sigma}\rangle = \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu]\,, \tag{1.48}$$

$$\langle \hat{s}^y_{\mathbf{k}\sigma}\rangle = 0\,, \tag{1.49}$$

$$\langle \hat{s}^z_{\mathbf{k}\sigma}\rangle = -\frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu]\,. \tag{1.50}$$

Magnetisation self-consistency equation. We gave the self-consistency equation for the magnetisation m in Eq. (1.19). By further elaboration, it can be written equivalently as

$$m = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta_{\mathbf{k}\uparrow}}{E_{\mathbf{k}\uparrow}}\mathrm{Th}^{(-)}_{\mathbf{k}\uparrow}[\beta,\mu]\,. \tag{1.51}$$

This equation is the "operational" version of the self-consistency equation for the HFP m.

Proof 1.10 — Proof of Eq. (1.51): self-consistency equation for m.

Consider Eq. (1.19). Restricting the sum to the MBZ and using Eq. (1.23), we get

$$m = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\left(\hat{\psi}_{\mathbf{k}\uparrow}+\hat{\psi}^\dagger_{\mathbf{k}\uparrow}\right)-\left(\hat{\psi}_{\mathbf{k}\downarrow}+\hat{\psi}^\dagger_{\mathbf{k}\downarrow}\right)\right\rangle\,.$$

From Eq. (1.37), we recognise a pseudospin EV in direction x. From Eq. (1.48)

$$\begin{aligned}
m &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left(\langle\hat{s}^x_{\mathbf{k}\uparrow}\rangle-\langle\hat{s}^x_{\mathbf{k}\downarrow}\rangle\right)\\
&= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left(\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\uparrow}\}}{E_{\mathbf{k}\uparrow}}\mathrm{Th}^{(-)}_{\mathbf{k}\uparrow}[\beta,\mu]-\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}\downarrow}}\mathrm{Th}^{(-)}_{\mathbf{k}\downarrow}[\beta,\mu]\right)\\
&= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta_{\mathbf{k}\uparrow}}{E_{\mathbf{k}\uparrow}}\mathrm{Th}^{(-)}_{\mathbf{k}\uparrow}[\beta,\mu]\,,
\end{aligned}$$

which proves Eq. (1.51). In last line we used $\Delta_{\mathbf{k}\uparrow}=-\Delta_{\mathbf{k}\downarrow}$, which follows from Eq. (1.27), and the fact that $E_{\mathbf{k}\sigma}$ and $\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu]$ are spin actually spin-independent. □

Density. The averaged density obeys the equations

$$n = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\mathrm{Th}^{(+)}_{\mathbf{k}\uparrow}[\beta,\mu] \tag{1.52}$$

$$= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\mathrm{Th}^{(+)}_{\mathbf{k}\uparrow}[\beta,\mu]\,. \tag{1.53}$$

The two are equivalent, but the second one is handier to implement in numeric computations.

Proof 1.11 — Proof of Eq. (1.52): self-consistency equation for n.

The $\mathcal{F}$-transform of the on-site density operator satisfies

$$\begin{aligned}\sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} \\ &\overset{\mathcal{F}}{=} \sum_{\mathbf{r}\in\mathcal{L}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}} \hat{c}_{\mathbf{k}'\sigma}\,,\end{aligned} \tag{1.54}$$

where we used Eq. (??). By further elaboration, we obtain

$$\begin{aligned}\sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\mathbf{k}'\in\mathrm{BZ}} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{-i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}} \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{n}_{\mathbf{k}\sigma}\,,\end{aligned} \tag{1.55}$$

where we used Eq. (??). Hence, applying Eq. (1.2),

$$\begin{aligned} n &= \frac{1}{2}\sum_\sigma \left(\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} n + \overbrace{\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y} m}^{=0} \right) \\ &= \frac{1}{2L_x L_y} \sum_\sigma \sum_{\mathbf{r}\in\mathcal{L}} \langle \hat{n}_{\mathbf{r}\sigma}\rangle \\ &= \frac{1}{2L_x L_y} \sum_\sigma \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{n}_{\mathbf{k}\sigma}\rangle\,.\end{aligned} \tag{1.56}$$

Restricting the sum of Eq. (1.56) to the MBZ, we get

$$\begin{aligned} n &= \frac{1}{2L_x L_y} \sum_\sigma \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle \\ &= \frac{1}{2L_x L_y} \sum_\sigma \sum_{\mathbf{k}\in\mathrm{MBZ}} \left\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma}\right\rangle\,.\end{aligned} \tag{1.57}$$

The mapping $\hat{\Psi} \to \hat{\Gamma}$ is unitary, so

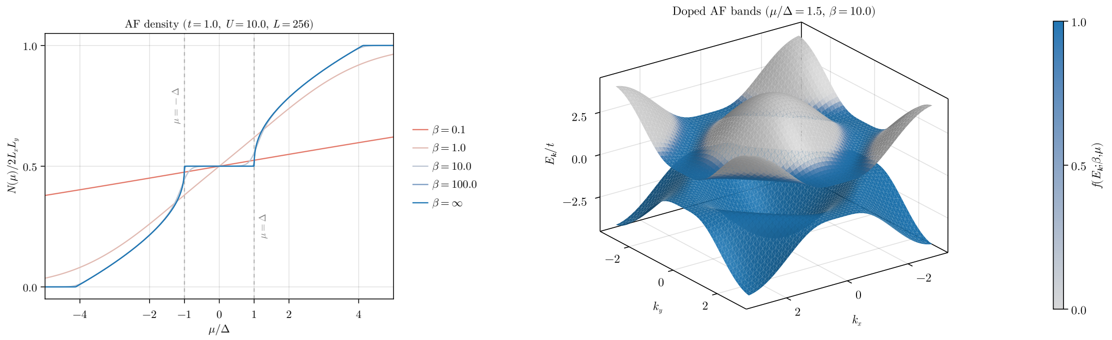
$$\hat{\Psi}^\dagger_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} = \hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\hat{\Gamma}_{\mathbf{k}\sigma}\,.$$

Therefore, Eq. (1.57) reduces to

$$\begin{aligned} n &= \frac{1}{2L_x L_y} \sum_\sigma \sum_{\mathbf{k}\in\mathrm{MBZ}} \left\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\hat{\Gamma}_{\mathbf{k}\sigma}\right\rangle \\ &= \frac{1}{2L_x L_y} \sum_\sigma \sum_{\mathbf{k}\in\mathrm{MBZ}} \left(\left\langle \hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(+)}_{\mathbf{k}\sigma}\right\rangle + \left\langle \hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(-)}_{\mathbf{k}\sigma}\right\rangle \right) \\ &= \frac{1}{2L_x L_y} \sum_\sigma \sum_{\mathbf{k}\in\mathrm{MBZ}} \mathrm{Th}^{(+)}_{\mathbf{k}\sigma}[\beta,\mu]\,.\end{aligned}$$

In last line we used Eq. (1.47). The thermal factor is spin-independent, hence we can set $\sigma = \uparrow$ and remove the sum over spin. This proves Eq. (1.52). □

In Fig. 1.3a we sketch density as a function of the chemical potential, computed using Eq. (1.52) at $U/t = 10$ and $L = 256$. As we increase β, we observe that a sort of plateau emerges at $\mu \in [-\Delta, \Delta]$, becoming increasingly sharper as we decrease temperature. For an insulating phase, any infinitesimal shift to μ does not increase n. Indeed, in that case the chemical potential lies



a | Electronic density. b | Example doped bands.

Figure 1.3 | Left panel: electronic density in the AF phase as a function of the chemical potential, computed through Eq. (1.53), at various temperatures. Right panel: example doped bands, with $\mu/\Delta = 1.5$ set arbitrarily and temperature set to $\beta = 10$ to enhance smearing effects around the FS. Single-state filling is proportional to the colour intensity.

within a gap and no states can be filled by incrementing a little the chemical potential. On the contrary, for a metallic phase n changes as we vary infinitesimally μ. In Fig. 1.3a, we observe that the plateau is located at $n = 1/2$, i.e. half-filling, while for $|\mu| > \Delta$ the density varies with μ. Therefore, the commensurate AF phase is insulating only at half-filling.

Instability of the doped commensurate antiferromagnet. Consider now the gran canonical hamiltonian $\hat{H} - \mu\hat{N}$ and use Eq. (1.40). The resulting operator is diagonalised by the same matrix given in Eq. (1.43), being

$$-\mu\hat{N} = \sum_{\mathbf{k}\in\mathrm{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \implies W_{\mathbf{k}\sigma}\left[h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times 2}\right] W^{\dagger}_{\mathbf{k}\sigma} = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2\times 2}\,, \tag{1.58}$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. These two bands have the same number of single-particle states. In Fig. 1.3b we sketch the bands $\pm E_{\mathbf{k}\sigma}$. Intense colour indicates an almost filled single-particle state, lighter gray colour an almost empty single-particle state. The gap is at zero energy, but at any finite doping $\tilde{\mu}$ crosses the upper band. In terms of band theory classifications, the resulting phase is metallic – except at half-filling, when the chemical potential equals zero and lies within the gap. In that case, the phase is insulating.

As we said at the beginning of this chapter, there exist several ways to realise an Hubbard antiferromagnet. For the conventional HM, it is known [26, 27] that the commensurate AF phase is unstable at any finite doping. This is discussed briefly in App. **??**. The key takeaway is that insulating antiferromagnets are stable with respect to spin fluctuations, while metallic antiferromagnets are not.

In fact, the mere existence of a self-consistent solution does not guarantee stability: indeed, when in Sec. **??** we derived the variational solution, we only required it to be a stationary point of the HF energy. There is no need for the solution to be a minimum. In particular, the commensurate AF phase is a saddle point of HF energy and it can be shown that any finite doping disrupts commensurate ordering towards uncommensurate phases [26].

1.3 Extension to the EHM

We now include the non-local interaction $\hat{H}_V$. We anticipate here that we are going to obtain the same HF hamiltonian of Eq. (1.26) with renormalised chemical potential, bands and gap,

$$\mu \to \tilde{\mu}\,, \qquad \epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma}\,, \qquad \Delta_{\mathbf{k}\sigma} \to \tilde{\Delta}_{\mathbf{k}\sigma}\,.$$

The explicit expressions for these renormalised quantities will be given, respectively, in Eqs. (1.72), (1.85) and (1.102). Due to this renormalisation of parameters, the diagonalisation procedure of Sec. 1.2.2 can be generalised.

In diagonal form, the bands $\pm\tilde{E}_{\mathbf{k}\sigma}$ read

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}\,.$$

We allow $\tilde{\Delta}_{\mathbf{k}\sigma}$ to be a complex function of $\mathbf{k}$ and σ,

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + i\,\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}\,.$$

We define the renormalised angle $\tilde{\theta}_{\mathbf{k}\sigma}$, $\tilde{\zeta}_{\mathbf{k}\sigma}$ through the relations:

$$\sin 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}}\,, \qquad \cos 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}}\,, \tag{1.59}$$

and

$$\sin 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|}\,, \qquad \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|}\,. \tag{1.60}$$

A sketch of these angles and the generalised diagonalisation procedure is given in Fig. 1.4. For a complex gap function, the new diagonalisation matrix is given by:

$$\begin{aligned}
\tilde{W}_{\mathbf{k}\sigma} &\equiv e^{-i\tilde{\theta}_{\mathbf{k}\sigma}\tau^y} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}\tau^z} \\
&= \left(\cos\tilde{\theta}_{\mathbf{k}\sigma} - i\sin\tilde{\theta}_{\mathbf{k}\sigma}\tau^y\right)\left(\cos\tilde{\zeta}_{\mathbf{k}\sigma} - i\sin\tilde{\zeta}_{\mathbf{k}\sigma}\tau^z\right) \\
&= \begin{bmatrix} \cos\tilde{\theta}_{\mathbf{k}\sigma} & -\sin\tilde{\theta}_{\mathbf{k}\sigma} \\ \sin\tilde{\theta}_{\mathbf{k}\sigma} & \cos\tilde{\theta}_{\mathbf{k}\sigma} \end{bmatrix} \begin{bmatrix} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & \\ & e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \end{bmatrix} \\
&= \begin{bmatrix} \cos\tilde{\theta}_{\mathbf{k}\sigma} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & -\sin\tilde{\theta}_{\mathbf{k}\sigma} e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \\ \sin\tilde{\theta}_{\mathbf{k}\sigma} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & \cos\tilde{\theta}_{\mathbf{k}\sigma} e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \end{bmatrix}\,,
\end{aligned}$$

The following relations hold (*tilde* sign indicates the renormalised quantity):

$$\begin{aligned}
\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^x\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= -\sin 2\tilde{\theta}_{\mathbf{k}\sigma}\cos 2\tilde{\zeta}_{\mathbf{k}\sigma}\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle\,, \\
\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^y\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= -\sin 2\tilde{\theta}_{\mathbf{k}\sigma}\sin 2\tilde{\zeta}_{\mathbf{k}\sigma}\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle\,, \\
\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= \cos 2\tilde{\theta}_{\mathbf{k}\sigma}\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle\,.
\end{aligned}$$

Let us extend Eq. (1.46) and define the renormalised thermal factors,

$$\langle\hat{\Gamma}^\dagger_{\mathbf{k}\sigma}\tau^z\hat{\Gamma}_{\mathbf{k}\sigma}\rangle = f\left(\tilde{E}_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right)\,, \tag{1.61}$$

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta,\tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right) \pm f\left(\tilde{E}_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right)\,, \tag{1.62}$$

At half-filling, we have:

$$\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}} = 1\,, \tag{1.63}$$

$$\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}} = \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\,. \tag{1.64}$$

Proof 1.12 — Proof of Eqs. (1.63), (1.64).

At half-filling, $\mu = 0$. For a null chemical potential, the Fermi-Dirac distribution is given by

$$f(\epsilon;\beta,0) = \frac{1}{e^{\beta\epsilon}+1}\,.$$

Therefore:

1. The following identity holds:

$$\begin{aligned} f(-\epsilon;\beta,0) &= \frac{1}{e^{-\beta\epsilon}+1} \\ &= \frac{e^{\beta\epsilon}}{1+e^{\beta\epsilon}} \\ &= 1-\frac{1}{e^{\beta\epsilon}+1} \\ &= 1-f(\epsilon;\beta,0)\,. \end{aligned}$$

   Hence,

$$f(-\tilde{E}_{\mathbf{k}};\beta,0)+f(\tilde{E}_{\mathbf{k}};\beta,0)=1$$

   which proves Eq. (1.63).

2. We have:

$$\begin{aligned} f(-\epsilon;\beta,0)-f(\epsilon;\beta,0) &= \frac{1}{e^{-\beta\epsilon}+1}-\frac{1}{e^{\beta\epsilon}+1} \\ &= \frac{e^{\beta\epsilon}}{e^{\beta\epsilon}+1}-\frac{1}{e^{\beta\epsilon}+1} \\ &= \frac{e^{\beta\epsilon/2}-e^{-\beta\epsilon/2}}{e^{\beta\epsilon/2}+e^{-\beta\epsilon/2}}\,. \end{aligned}$$

   Therefore,

$$f(-\tilde{E}_{\mathbf{k}};\beta,0)-f(\tilde{E}_{\mathbf{k}};\beta,0)=\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$$

   which proves Eq. (1.64).

□

Pseudospin EVs in the three directions are:

$$\langle \hat{s}^x_{\mathbf{k}\sigma}\rangle = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{\tilde{E}_{\mathbf{k}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta,\tilde{\mu}]\,, \tag{1.65}$$

$$\langle \hat{s}^y_{\mathbf{k}\sigma}\rangle = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{\tilde{E}_{\mathbf{k}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta,\tilde{\mu}]\,, \tag{1.66}$$

$$\langle \hat{s}^z_{\mathbf{k}\sigma}\rangle = -\frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta,\tilde{\mu}]\,. \tag{1.67}$$

These equations are an extension of Eqs. (1.48), (1.49), (1.50) for the EHM.

By renormalising all quantities, the self-consistency equation for magnetisation must stay formally the same of Eq. (1.51),

$$m=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\tilde{\Delta}_{\mathbf{k}\uparrow}}{\tilde{E}_{\mathbf{k}\uparrow}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\uparrow}[\beta,\tilde{\mu}]\,. \tag{1.68}$$

We will be able to prove this statement rigorously in Sec. 1.4. For the density n an identical argument holds, and Eq. (1.52) is extended to

$$n=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}\uparrow}[\beta,\tilde{\mu}]\,. \tag{1.69}$$

With this strategy, we get back to the application of HF methods to the full EHM, including the non-local attraction.

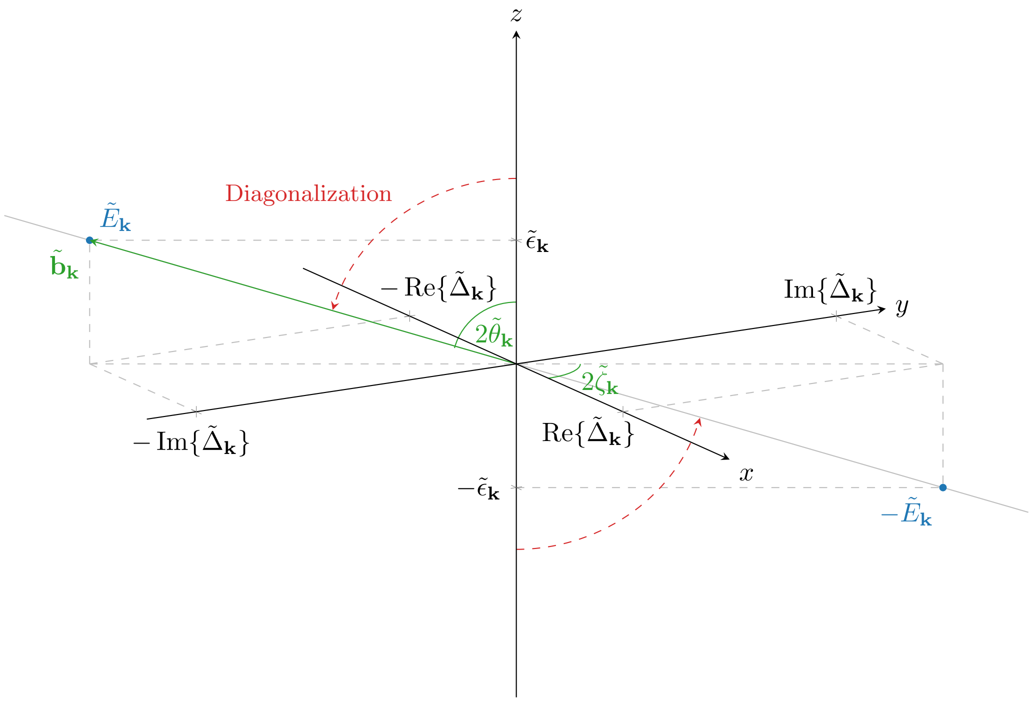


Figure 1.4 Sketch of the diagonalisation procedure for the EHM antiferromagnet. To diagonalise the HF hamiltonian, we align the z axis with the pseudo-field. This is an extesion of Fig. 1.2.

1.3.1 Effects of the non-local Hartree terms: chemical potential renormalisation

The approximation to $\hat{H}_V$ in the non-local Hartree channel is given by

$$\hat{H}_V \overset{\text{HF}}{\simeq} -2znV \times \hat{N} - E^{(\text{AF})}_{\text{H},V} + (\text{Fock channel})\,, \tag{1.70}$$

with the contraction energy

$$E^{(\text{AF})}_{\text{H},V} \equiv V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma,\sigma'} \big\langle \hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma} \big\rangle \big\langle \hat{c}^{\dagger}_{\mathbf{r}'\sigma'} \hat{c}_{\mathbf{r}'\sigma'} \big\rangle\,. \tag{1.71}$$

From Eq. (1.70), we see that the Hartree channel of the non-local attraction contributes to the chemical potential shift. Combining Eq. (1.26) with Eq. (1.70), the full shift to chemical potential reads

$$\tilde{\mu} \equiv \mu + n(2zV - U)\,, \tag{1.72}$$

which is identical to what we found for the normal phase, in Eq. (**??**).

Proof 1.13 — Proof of Eq. (1.70).

In Eq. (**??**) we separated the Hartree channel approximation to $\hat{H}_V$ in s.s. and o.s. channels. The same equation is valid in this context. Using the definition of $E^{(\text{AF})}_{\text{H},V}$ given in Eq. (1.71), we obtain

$$\begin{aligned}\hat{H}_V \overset{\text{HF}}{\simeq} &-V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left(\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\sigma} \rangle \right) \\ &- V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left(\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\overline{\sigma}} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle \right) - E^{(\text{AF})}_{\text{H},V} + (\text{Fock channel})\,. \end{aligned} \tag{1.73}$$

Inserting the Ansatz of Eq. (1.2), we obtain

$$\hat{H}_V \overset{\text{HF}}{\simeq} \overbrace{-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(\hat{n}_{\mathbf{r}'\sigma}+\hat{n}_{\mathbf{r}\sigma})+mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{\delta_{\sigma=\uparrow}}\left((-1)^{x+y}\hat{n}_{\mathbf{r}'\sigma}+(-1)^{x'+y'}\hat{n}_{\mathbf{r}\sigma}\right)}^{\text{s.s.}}$$
$$\underbrace{-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}'\bar{\sigma}}+\hat{n}_{\mathbf{r}\sigma}]+mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left((-1)^{x+y+\delta_{\sigma=\uparrow}}\hat{n}_{\mathbf{r}'\bar{\sigma}}+(-1)^{x'+y'+\delta_{\bar{\sigma}=\uparrow}}\hat{n}_{\mathbf{r}\sigma}\right)}_{\text{o.s.}}$$
$$-E^{(\text{AF})}_{\text{H},V}+(\text{Fock channel})\,.$$

For a square lattice, if $\mathbf{r}=(x,y)$ and $\mathbf{r}'=(x',y')$ are NNs, both the following identities are true

$$(-1)^{x'+y'}=(-1)^{x+y+1}\,,$$
$$(-1)^{\delta_{\bar{\sigma}=\uparrow}}=(-1)^{\delta_{\sigma=\uparrow}+1}\,.$$

We obtain

$$\hat{H}_V \overset{\text{HF}}{\simeq} \overbrace{-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma})}^{\text{s.s. (density)}}\overbrace{-mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}(\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma})}^{\text{s.s. (magnetisation)}}$$
$$\underbrace{-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}})}_{\text{o.s. (density)}}\underbrace{+mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}})}_{\text{o.s. (magnetisation)}}$$
$$-E^{(\text{AF})}_{\text{H},V}+(\text{Fock channel})\,. \tag{1.74}$$

In the above expressions we separated in “density” and “magnetisation” terms. Let us deal with them separately.

Density terms. In the sum of s.s. and o.s. density terms of Eq. (1.74) we recognise the operator of Eq. (**??**), obtained for the normal phase. Hence we obtain the same result of Eq. (**??**),

$$-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma})-nV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}})=-2znV\times\hat{N}\,. \tag{1.75}$$

Magnetisation terms. The magnetisation s.s. and o.s. terms of Expr. (1.74) perfectly cancel out:

$$mV\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[(\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma})-(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}})\right]=0\,.$$

Therefore the only contribution to the HF approximation of $\hat{H}_V$, in the Hartree channel, comes from Eq. (1.75). Inserting the latter in Eq. (1.74), Eq. (1.70) is proven. □

The contraction energy of Eq. (1.71) reduces to

$$E^{(\text{AF})}_{\text{H},V}=-L_xL_y\times 2n^2\,, \tag{1.76}$$

which coincides with that for the normal phase, given in Eq. (**??**). This is coherent with the fact that the shift to chemical potential is identical.

Proof 1.14 — Proof of Eq. (1.76): non-local Hartree contraction energy.

Separate Eq. (1.71) in s.s. and o.s. sectors,

$$E^{(\mathrm{AF})}_{\mathrm{H},V} = V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \Big(\overbrace{\langle \hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma} \rangle \langle \hat{c}^{\dagger}_{\mathbf{r}'\sigma} \hat{c}_{\mathbf{r}'\sigma} \rangle}^{\text{s.s.}} + \overbrace{\langle \hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma} \rangle \langle \hat{c}^{\dagger}_{\mathbf{r}'\overline{\sigma}} \hat{c}_{\mathbf{r}'\overline{\sigma}} \rangle}^{\text{o.s.}} \Big) . \tag{1.77}$$

s.s. sector contraction energy. The s.s. sector of Eq. (1.77) reduces to

$$V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle = \overbrace{V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \Big[(n + (-1)^{x+y} m)(n + (-1)^{x'+y'} m) \Big]}^{\sigma=\uparrow}$$
$$+ \underbrace{V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \Big[(n - (-1)^{x+y} m)(n - (-1)^{x'+y'} m) \Big]}_{\sigma=\downarrow} .$$

The mixed terms (those of order $\mathcal{O}(n) \times \mathcal{O}(m)$) cancel out, and since sign factors necessarily alternate on neighbouring sites, the remainder is just

$$V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} 2(n^2 - m^2) . \tag{1.78}$$

o.s. sector contraction energy. Proceeding identically on the o.s. sector of Eq. (1.77), we get

$$V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle = \overbrace{V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \Big[(n + (-1)^{x+y} m)(n - (-1)^{x'+y'} m) \Big]}^{\sigma=\uparrow}$$
$$+ \underbrace{V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \Big[(n - (-1)^{x+y} m)(n + (-1)^{x'+y'} m) \Big]}_{\sigma=\downarrow} ,$$

which gives

$$V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} 2(n^2 + m^2) . \tag{1.79}$$

Then, summing Eqs. (1.78) and (1.79), we get

$$E^{(\mathrm{AF})}_{\mathrm{H},V} = -2V \sum_{\langle ij\rangle} \sum_{\sigma} n^2 = -4n^2 V \times \frac{z}{2} L_x L_y .$$

which coincides with Eq. (1.76). □

1.3.2 Effects of the Fock terms: gap and bare bands renormalisation

In this subsection we discuss the Fock contributions to the HF approximated hamiltonian, in the AF phase. We start by Eq. (**??**), derived in the context of the normal phase,

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel})$$
$$+ \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} (\cos(\delta k_x) + \cos(\delta k_y)) \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E^{(\mathrm{AF})}_{\mathrm{F},V} , \tag{1.80}$$

where $E^{(\mathrm{AF})}_{\mathrm{F},V}$, see Eq. (**??**), reads

$$E^{(\mathrm{AF})}_{\mathrm{F},V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left(\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right) \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \,, \tag{1.81}$$

As we specified in Sec. 1.1.2, the commensurate AF phase is described by two EVs being in non-zero in reciprocal space,

$$\left|\langle \hat{n}_{\mathbf{k}\sigma} \rangle\right| \geq 0 \quad \text{and} \quad \left|\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle\right| \geq 0\,,$$

This implies only two contributions in Eq. (1.80) are non-zero:

$$\mathbf{k} = -\mathbf{k}' \quad \text{and} \quad \mathbf{k} + \boldsymbol{\pi} = -\mathbf{k}'\,.$$

Then Eq. (1.80) becomes

$$\begin{aligned} \hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) &+ \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos\left(2k_x\right) + \cos\left(2k_y\right)\right] \\ &\times \Big(\underbrace{\langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}} - \underbrace{\langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}} \Big) - E^{(\mathrm{AF})}_{\mathrm{F},V}\,, \end{aligned} \tag{1.82}$$

where we separated the hamiltonian in *diagonal* and *off-diagonal* terms.

Diagonal terms. The diagonal terms of Eq. (1.82) reduce to

$$\begin{aligned} &\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos\left(2k_x\right) + \cos\left(2k_y\right)\right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ &\qquad = V \sum_{\mathbf{k},\sigma} \left(u^{(s^*)}_{\sigma} (\cos q_x + \cos q_y) + u^{(d)}_{\sigma} (\cos q_x - \cos q_y) \right) \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}\,, \end{aligned} \tag{1.83}$$

where we used Eq. (1.12), where the harmonics of $u_{\mathbf{k}\sigma}$ were defined.

Proof 1.15 — Proof of Eq. (1.83).

We define new variables as in Fig. **??**, $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. Applying this change of variables to the diagonal terms of Eq. (1.82), we obtain

$$\begin{aligned} &\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos\left(2k_x\right) + \cos\left(2k_y\right)\right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ &\qquad = \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos\left(\delta q_x\right) + \cos\left(\delta q_y\right)\right] \langle \hat{c}^{\dagger}_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \\ &\qquad = \sum_{\gamma,\sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \right) \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{n}_{\mathbf{q}'\sigma} \\ &\qquad = \sum_{\gamma,\sigma} u^{(\gamma)}_{\sigma} \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{n}_{\mathbf{q}'\sigma}\,. \end{aligned} \tag{1.84}$$

We used the decomposition of Eq. (**??**). In the last line, we recognised the harmonics given in Eq. (1.12).

We can re-use the argument that, for the normal phase, lead us to Eq. (**??**). We are not breaking inversion symmetry (and as a consequence, neither C_2): this implies that, whatever the shape of the renormalised bands, it must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \quad \Longrightarrow \quad u^{(p_\ell)}_{\sigma} = 0 \quad \text{with} \quad \ell \in \{x, y\}\,.$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}]$	Fig. ??
$d_{x^2-y^2}$-wave	$u^{(d)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}]$	Fig. ??

Table 1.2 | Symmetry and spin resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

Renormalised bare bands need to be symmetric functions of the moment. Hence for $\gamma \in \{p_x, p_y\}$ the above sum vanishes. Using this selection rule, Eq. (1.90) reduces to Eq. (1.83). □

Apart from the explicit form of the self-consistency equations, this discussion is formally the same of the normal phase, carried out in Sec. ??. Therefore the bands renormalisation, discussed in Sec. ??, applies also for the AF phase. Specifically, the renormalised bands are

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u^{(s^*)}_\sigma V(\cos k_x + \cos k_y) + u^{(d)}_\sigma V(\cos k_x - \cos k_y)\,. \tag{1.85}$$

The self-consistency equation for the harmonics of $u_{\mathbf{k}\sigma}$, given in Eq. (1.12), can be written as

$$u^{(\gamma)}_\sigma = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}]\,. \tag{1.86}$$

The symmetry-resolved self-consistency equations are reported explicitly in Tab. 1.3. Since in Eq. (1.86) all quantities are spin-independent, we omit spin index on these HFPs.

Proof 1.16 — Proof of Eq. (1.86): self-consistency equations for $u^{(\gamma)}_\sigma$.

Recall the definition of $u_{\mathbf{k}\sigma}$,

$$u_{\mathbf{k}\sigma} = \langle \hat{n}_{\mathbf{k}\sigma} \rangle\,,$$

given in Eq. (1.10). We expand the harmonics decomposition of Eq. (1.12),

$$\begin{aligned} u^{(\gamma)}_\sigma &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left(\varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{n}_{\mathbf{k}\sigma} \rangle + \varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} \langle \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \right) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle\,. \end{aligned}$$

In last line we used Eq. (1.24). From Eq. (1.39), we recognise the z component of the pseudospin operator

$$\begin{aligned} u^{(\gamma)}_\sigma &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{s}^z_{\mathbf{k}\sigma} \rangle \\ &= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}]\,, \end{aligned} \tag{1.87}$$

where, in the last line, Eq. (1.67) has been used. This proves Eq. (1.86). □

Similarly to the normal phase, also for the AF phase we can define the symmetry-resolved components of the renormalised hopping amplitude,

$$\tilde{t}^{(s^*)}_\sigma \equiv t - \frac{u^{(s^*)}_\sigma V}{2} \quad \text{and} \quad \tilde{t}^{(d)}_\sigma \equiv -\frac{u^{(d)}_\sigma V}{2}\,. \tag{1.88}$$

We re-used the definitions of Eq. (??).

Off-diagonal terms. The off-diagonal sector of Eq. (1.82) is reduced to

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= V\sum_{\gamma,\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left(iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{k}}\hat{\psi}_{\mathbf{k}\sigma}+\mathrm{h.c.}\right), \quad (1.89)$$

where we used the harmonics of $v_{\mathbf{k}\sigma}$, defined in Eq. (1.13).

Proof 1.17 — Proof of Eq. (1.89).

We define new variables as in Fig. **??**, $\mathbf{q}\equiv\mathbf{K}+\mathbf{k}$, $\mathbf{q}'\equiv\mathbf{K}-\mathbf{k}$. Applying this change of variables to the off-diagonal terms of Eq. (1.82), we obtain

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$=-\sum_{\gamma,\sigma}\left(\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}^\dagger_{\mathbf{q}\sigma}\rangle\right)\times V\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{\psi}^\dagger_{\mathbf{q}'\sigma}. \quad (1.90)$$

We shift the first sum by an AF vector $\boldsymbol{\pi}$ defining $\mathbf{k}+\boldsymbol{\pi}\equiv\mathbf{q}$, and use Eq. (1.22) as well as reciprocal space periodicity by a BZ,

$$\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}^\dagger_{\mathbf{q}\sigma}\rangle=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}\langle\hat{\psi}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle$$
$$=-\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle.$$

Recall the definition $v_{\mathbf{k}\sigma}=i\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle$ of Eq. (1.11). Inserting the above equation in Eq. (1.90), we obtain

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$=\sum_{\gamma,\sigma}\left(\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle\right)\times V\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{\psi}^\dagger_{\mathbf{q}'\sigma}$$
$$=-i\sum_{\gamma,\sigma}v^{(\gamma)}_\sigma\times V\sum_{\mathbf{q}'}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{n}_{\mathbf{q}'\sigma}, \quad (1.91)$$

where we used Eq. (1.13). We restrict the sum to the MBZ,

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$=-iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left(v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}^\dagger_{\mathbf{q}\sigma}+v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}+\boldsymbol{\pi}}\hat{\psi}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma}\right),$$

and use Eq. (1.22), exchanging terms positions

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$=iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left(v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}-v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}^\dagger_{\mathbf{q}\sigma}\right). \quad (1.92)$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$v^{(s^*)} = \dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \dfrac{\mathrm{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_\mathbf{k}[\beta, \tilde{\mu}]$	Fig. ??
p_x-wave	$v^{(p_x)} = \dfrac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \sin k_x \dfrac{\mathrm{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_\mathbf{k}[\beta, \tilde{\mu}]$	Fig. ??
p_y-wave	$v^{(p_y)} = \dfrac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \sin k_y \dfrac{\mathrm{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_\mathbf{k}[\beta, \tilde{\mu}]$	Fig. ??
$d_{x^2-y^2}$-wave	$v^{(d)} = \dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x - \cos k_y) \dfrac{\mathrm{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_\mathbf{k}[\beta, \tilde{\mu}]$	Fig. ??

Table 1.3 | Symmetry and spin resolved self-consistency equations for the harmonics of $v_\mathbf{k}$.

Recalling Tab. ??, for the four symmetries we are considering, the functions $\varphi^{(\gamma)}_\mathbf{k}$ are real-valued for $\gamma = s^*, d$, and purely imaginary for $\gamma = p_x, p_y$. We anticipate the result of Eq. (1.95),

$$v^{(s^*)}_\sigma, v^{(d)}_\sigma \in \mathbb{R}\,, \qquad v^{(p_x)}_\sigma, v^{(p_y)}_\sigma \in i\mathbb{R}\,,$$

that will be obtained later. Then

$$i v^{(\gamma)}_\sigma \varphi^{(\gamma)*}_\mathbf{q} \in i\mathbb{R}\,, \tag{1.93}$$

and to conjugate a purely imaginary number it suffices to put a minus sign in front of it,

$$-i v^{(\gamma)}_\sigma \varphi^{(\gamma)*}_\mathbf{q} = \left(i v^{(\gamma)}_\sigma \varphi^{(\gamma)*}_\mathbf{q} \right)^*\,.$$

Therefore, substituting this identity in Eq. (1.92), we obtain Eq. (1.89). □

The HFPs $v^{(\gamma)}_\sigma$ obey the self-consistency equations

$$\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_\mathbf{q} \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}]\,. \tag{1.94}$$

These equation are reported, symmetry-resolved, in Tab. 1.4. By direct inspection of the equations thereby reported, we see that

$$v^{(s^*)}_\sigma, v^{(d)}_\sigma \in \mathbb{R}\,, \qquad v^{(p_x)}_\sigma, v^{(p_y)}_\sigma \in i\mathbb{R}\,. \tag{1.95}$$

We have used this result before, in Eq. (1.93).

In Tab. 1.4 the four derived self-consistency equations are reported, ignoring spin index. Indeed, all quantities in Eq. (1.94) are independent of σ.

Proof 1.18 — Proof of Eq. (1.94): self-consistency equations for $v^{(\gamma)}_\sigma$.

We expand Eq. (1.13),

$$\begin{aligned} v^{(\gamma)}_\sigma &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_\mathbf{q} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \\ &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left(\varphi^{(\gamma)}_\mathbf{q} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} \langle \hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right) \\ &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_\mathbf{q} \langle \hat{\psi}_{\mathbf{q}\sigma} - \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle\,. \end{aligned}$$

In last line we used Eq. (1.24). From Eq. (1.38), we recognise the y component of the pseudospin operator

$$v_\sigma^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma} \tau^y \hat{\Psi}_{\mathbf{q}\sigma} \rangle$$
$$= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\text{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}] \,, \tag{1.96}$$

where we used Eq. (1.66). Last line coincides with Eq. (1.94). □

The contraction energy of Eq. (1.81) reduces to

$$E_{\text{F},V}^{(\text{AF})} = L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left(\left| u_\sigma^{(\gamma)} \right|^2 + \left| v_\sigma^{(\gamma)} \right|^2 \right) \,, \tag{1.97}$$

Note that, due to spin independency, we can omit the spin index, remove the sum over σ and the 1/2 prefactor. The new HFPs amplitudes contribute positively to contraction energy. The consequences of this fact will be discussed in Sec. 1.5.

Proof 1.19 — Proof of Eq. (1.97): Fock contraction energy.

Consider Eq. (1.81). Using the changing variables of Fig. **??**, we get

$$E_{\text{F},V}^{(\text{AF})} = \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] \Big(\underbrace{\langle \hat{n}_{\mathbf{q}\sigma} \rangle \langle \hat{n}_{\mathbf{q}'\sigma} \rangle}_{\text{Diagonal terms}} - \underbrace{\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \langle \hat{\psi}_{\mathbf{q}'\sigma} \rangle}_{\text{Off-diagonal terms}} \Big) \,.$$

Applying the decomposition of Eq. (**??**) and using Eqs. (1.10), (1.11), we obtain:

$$E_{\text{F},V}^{(\text{AF})} = \underbrace{\frac{V}{2L_x L_y} \sum_{\gamma,\sigma} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma}}_{\text{Diagonal terms}} + \underbrace{\frac{V}{2L_x L_y} \sum_{\gamma,\sigma} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma}}_{\text{Off-diagonal terms}} \,. \tag{1.98}$$

Diagonal terms. Let us start from the diagonal part of Eq. (1.98). It holds

$$u_{\mathbf{q}\sigma} = \langle \hat{n}_{\mathbf{q}\sigma} \rangle \quad \Longrightarrow \quad u_{\mathbf{q}\sigma} = u^*_{\mathbf{q}\sigma} \,,$$

and then

$$\frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} = \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right) \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u^*_{\mathbf{q}'\sigma} \right)$$
$$= \left| u_\sigma^{(\gamma)} \right|^2 . \tag{1.99}$$

Off-diagonal terms. Consider the off-diagonal part of Eq. (1.98). Using Eq. (1.23), we get

$$v_{\mathbf{q}\sigma} = i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \quad \Longrightarrow \quad v_{\mathbf{q}+\boldsymbol{\pi}\sigma} = -v^*_{\mathbf{q}\sigma} \,,$$

Since the form factors satisfy $\varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)} = -\varphi_{\mathbf{q}}^{(\gamma)}$, we get:

$$\frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} = \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} v_{\mathbf{q}\sigma} \right) \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}'\sigma} \right)$$
$$= v_\sigma^{(\gamma)} \left(\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)*} v_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right) \,,$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k} + \boldsymbol{\pi}$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in \mathrm{BZ}$.

$$\begin{aligned}\frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi^{(\gamma)}_{\mathbf{q}} \varphi^{(\gamma)*}_{\mathbf{q}'} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} &= v^{(\gamma)}_{\sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{k}} v^{*}_{\mathbf{k}\sigma} \right) \\ &= \left| v^{(\gamma)}_{\sigma} \right|^2 . \end{aligned} \tag{1.100}$$

Inserting Eqs. (1.99), (1.100) in Eq. (1.98), we obtain the expression of Eq. (1.97). □

We examined all contribution to the HF hamiltonian. In next section we collect all results and describe the features of the HF ground-state.

1.4 The HF hamiltonian and its properties

We recollect all results into a single hamiltonian approximation. Summing Eqs. (1.26), (1.83) and (1.89), we obtain the HF hamiltonian

$$\begin{aligned}\hat{H} - \mu \hat{N} \overset{\mathrm{HF}}{\simeq} &\sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} \left(\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right) \\ &- \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left(\tilde{\Delta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}^{*}_{\mathbf{k}\sigma} \hat{\psi}^{\dagger}_{\mathbf{k}\sigma} \right) - \tilde{\mu} \hat{N} + E^{(\mathrm{AF})}_{\mathrm{c}} . \end{aligned} \tag{1.101}$$

We defined renormalised bands, see Eq. (1.85), as

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma\in\{s^*,d\}} u^{(\gamma)}_{\sigma} \varphi^{(\gamma)*}_{\mathbf{k}} ,$$

the renormalised gap function as

$$\tilde{\Delta}_{\mathbf{k}\sigma} \equiv \Delta_{\mathbf{k}\sigma} - iV \sum_{\gamma} v^{(\gamma)}_{\sigma} \varphi^{(\gamma)*}_{\mathbf{k}} , \tag{1.102}$$

and total contraction energy as the (negative) sum of all contraction energies, see Eqs. (1.31), (1.76) and (1.97),

$$E^{(\mathrm{AF})}_{\mathrm{c}} \equiv - \left(E^{(\mathrm{AF})}_{\mathrm{H},U} + E^{(\mathrm{AF})}_{\mathrm{H},V} + E^{(\mathrm{AF})}_{\mathrm{F},V} \right) . \tag{1.103}$$

1.4.1 Gap complexification and AF flavours

Recalling Eq. (1.93), we observe that the correction to the gap function in Eq. (1.102) is purely imaginary. Instead, the HM gap is purely real, $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU \in \mathbb{R}$. We denote this effect as "gap complexification". A direct consequence of the fact that the correction is purely imaginary is that Eq. (1.68), which expresses the self-consistency condition for m, which we postulated, is proven.

Proof 1.20 — Proof of Eq. (1.68): self-consistency equation for m.

We know from Eq. (1.65) that the real part of the gap is connected to the x projection of the pseudospin expectation value. Instead, from Eq. (1.66) we know that the imaginary part is related to the y projection. It follows

$$\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = (-1)^{\delta_{\sigma=\downarrow}} mU \in \mathbb{R}$$

Then, the x projection of the pseudospin gives a spin-antisymmetric EV,

$$\left\langle \hat{s}^{x}_{\mathbf{k}\uparrow} \right\rangle = - \left\langle \hat{s}^{x}_{\mathbf{k}\downarrow} \right\rangle$$

Phase	HFPs
sAF	$\{m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)}\}$
aAF	$\{m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 1.4 Set of HFP for the two possible realisations of the AF phase, respectively with a symmetric and an antisymmetric imaginary part of the gap.

Then Eq. (1.68) holds identically, because the proof that lead us to Eq. (1.51) for the HM was based on this property. □

In this analysis, antiferromagnetism is realised without breaking inversion symmetry. This translates in the requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \stackrel{!}{=} \tilde{E}_{-\mathbf{k}\sigma}\,.$$

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + \mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}\,.$$

Renormalised bare bands as well as the real part of the gap satisfy said inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \qquad \mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \mathrm{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}\,,$$

hence inversion symmetry is respected only if

$$\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm\,\mathrm{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}\,.$$

Either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this consideration, we are able two discard two of the four self-consistency equations for the harmonics $v^{(\gamma)}_\sigma$, listed in Tab. 1.4. If $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is an even function of momentum, $v^{(p_x)}_\sigma = v^{(p_y)}_\sigma = 0$. If $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is an odd function of momentum, $v^{(s^*)}_\sigma = v^{(d)}_\sigma = 0$. The two options cannot coexist, and separate simulations can be implemented in the two symmetry sectors.

We name the two possible symmetries "sAF" (symmetric antiferromagnet) and "aAF" (antisymmetric antiferromagnet). The set of HFPs then restricts to these AF structures, and is reported for both cases in Tab. 1.5. As we will discuss in Sec. 1.5, the two phases actually coincide.

1.4.2 The entropy density of the AF ground-state

The approximated ground-state in the AF phase is given by a gas of free $\hat{\gamma}^{(\pm)}$ fermions populating the two bands $\pm\tilde{E}_{\mathbf{k}}$ following a Fermi-Dirac distribution. Hence, the total entropy density of the ground state is given by that of a Fermi gas [4]:

$$s^{(\mathrm{AF})} = s^{(\mathrm{AF})}_{-} + s^{(\mathrm{AF})}_{+}\,,$$

where the superscript indicates the band,

$$s^{(\mathrm{AF})}_{\pm} \equiv -\frac{k_B}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\Big[\Big(1-f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big)\ln\Big(1-f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\,. \tag{1.104}$$

Using Eq. (??), we can give an alternate formula for the entropy density

$$s^{(\mathrm{AF})} = -\frac{k_\mathrm{B}}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) - \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\tilde{E}_{\mathbf{k}}-\tilde{\mu})f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) - \frac{k_\mathrm{B}}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\ln\Big(1-f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\tilde{E}_{\mathbf{k}}+\tilde{\mu})f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\,. \tag{1.105}$$

The proof is identical to the one provided for the normal phase in Sec. **??**. Note that, as the gap closes the above expression reduces to the entropy density for the normal phase reported in Eq. (**??**). Indeed, we are summing over the MBZ, and the bare bands $\tilde{\epsilon}_{\mathbf{k}}$ change sign when crossing its boundary. Thus, substituting $\tilde{E}_{\mathbf{k}} \to \tilde{\epsilon}_{\mathbf{k}}$, the first term coincides with half the sum of Eq. (**??**), the second term with the other half.

1.4.3 The free energy density of the AF ground-state

We compute the density of free energy, following the same procedure we used for the normal phase in Sec. **??**. The free energy density is given by

$$f^{(\mathrm{AF})} \equiv \frac{F^{(\mathrm{AF})}}{L_x L_y}\,, \tag{1.106}$$

being $F^{(\mathrm{AF})}$ the total free energy,

$$F^{(\mathrm{AF})} = E_{\mathrm{HF}}^{(\mathrm{AF})} - \frac{S^{(\mathrm{AF})}}{k_{\mathrm{B}}\beta}\,, \tag{1.107}$$

and $S^{(\mathrm{AF})} \equiv L_x L_y \times s^{(\mathrm{AF})}$ the total entropy. Here $E_{\mathrm{HF}}^{(\mathrm{AF})}$ is the ground-state energy for the HF hamiltonian. In the AF phase, we compute $f^{(\mathrm{AF})}$ using the following formula:

$$\begin{aligned} f^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[\ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \ln\Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big)\Big] \\ + Um^2 - V\sum_{\gamma}\Big(\big|u^{(\gamma)}\big|^2 + \big|v^{(\gamma)}\big|^2\Big) + \tilde{\mu} n\,. \end{aligned} \tag{1.108}$$

In this expression, we see that $u^{(\gamma)}$s, $v^{(\gamma)}$s contribution is negative, while magnetisation contributes positively. At a first glance, this suggests that a system with no magnetisation and non-zero $u^{(\gamma)}$s, $v^{(\gamma)}$s would be energetically favoured. In that situation we would have a purely imaginary gap for a non-magnetised phase, which would be not so intuitive to explain physically. In next section we explain how the above expression for the free energy density can be used to give a constraint that proves that said situation is impossible. HFPs are interrelated and the physical solution does not admit arbitrary values.

Proof 1.21 — Proof of Eq. (1.108): free energy density for the AF phase.

Using Eqs. (1.106), (1.107), we have

$$f^{(\mathrm{AF})} = \frac{E_{\mathrm{HF}}^{(\mathrm{AF})}}{L_x L_y} - \frac{s^{(\mathrm{AF})}}{k_{\mathrm{B}}\beta}\,,$$

with $s^{(\mathrm{AF})}$ the entropy density computed in Eq. (1.105). Then, $f^{(\mathrm{AF})}$ is obtained by summing the following terms:

1. Quasiparticles energy, taking care of spin degeneracy,

$$e_{\mathrm{g}}^{(\mathrm{AF})} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\,.$$

2. Contraction energy,

$$\begin{aligned} e_{\mathrm{c}}^{(\mathrm{AF})} &= \frac{E_{\mathrm{c}}^{(\mathrm{AF})}}{L_x L_y} \\ &= U(m^2 - n^2) + V\left[2zn^2 - \sum_{\gamma}\Big(\big|u^{(\gamma)}\big|^2 + \big|v^{(\gamma)}\big|^2\Big)\right]. \end{aligned}$$

The total contraction energy $E_{\mathrm{c}}^{(\mathrm{AF})}$ was given in Eq. (1.103). The above result was obtained by inserting Eqbs. (1.31), (1.76) and (1.97).

3. Entropic energy,

$$e_s^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \,,$$

obtained by multiplying Eq. (1.105) by $-2/\beta k_{\mathrm{B}}$.

4. Chemical potential correction

$$e_n^{(\mathrm{AF})} = \mu n \,.$$

Composing everything,

$$f^{(\mathrm{AF})} \equiv e_{\mathrm{g}}^{(\mathrm{AF})} + e_{\mathrm{c}}^{(\mathrm{AF})} + e_s^{(\mathrm{AF})} + e_n^{(\mathrm{AF})} \,,$$

we obtain Eq. (1.108).

□

1.5 Analysis of the solution

In this section, we present some constraints on the solution for the AF phase, showing that gap complexification is absent. Then we discuss how the renormalisation of bands can induce magnetisation. In the last paragraph, we use these constraints to approximate and interpret the formula for free energy of Eq. (1.108).

Self-consistency constraints. In Tab. 1.5 we collected the two sets HFPs for the two flavours of the AF phase. Each set contains five parameters. The scenario appears quite complicated, but it is possible to show that

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \,, \tag{1.109}$$

and

$$\forall \gamma \quad : \quad v^{(\gamma)} = 0 \,. \tag{1.110}$$

The proof of Eqs. (1.109) and (1.110) is given in App. ??. As a consequence, there is no need to distinguish two AF phases,

$$\mathrm{sAF} = \mathrm{aAF} \,.$$

It follows that the AF gap coincides with the one given by Eq. (1.27),

$$\tilde{\Delta}_{\mathbf{k}} = mU$$

Therefore, the Bogoliubov bands are given by

$$\tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + (Um)^2} \geq Um \,. \tag{1.111}$$

At any finite magnetisation m, the bands are gapped.

Mott localisation-enhanced magnetisation. The AF phase is described by two HFPs: $u^{(s^*)}$ and m. Moreover, we know that the HFP $u^{(s^*)}$ takes part in the renormalisation of the hopping amplitude,

$$\tilde{t}^{(s^*)} = t - \frac{u^{(s^*)} V}{2} \,,$$

where we used Eq. (1.88). From the constraint of Eq. (1.109), we see that the hopping parameter is reduced in amplitude,

$$t \quad \rightarrow \quad 0 < \tilde{t}^{(s^*)} < t \,.$$

Bands are effectively shrunk by the non-local attraction, as happened for the normal phase.

From the analysis of antiferromagnetism in the Hubbard model, we know that the local repulsion $\hat{H}_U$ induces magnetisation. We now show that the bands shrinking enhances it. Consider the Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ given in Eq. (1.44),

$$\tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + |mU|^2} \,.$$

If $mU \gg \tilde{t}^{(s^*)}$, these are approximated by

$$\begin{aligned} \tilde{E}_{\mathbf{k}} &= |mU| \sqrt{1 + \left(\frac{\tilde{\epsilon}_{\mathbf{k}}}{mU}\right)^2} \\ &\simeq mU \left[1 + \frac{1}{2}\left|\frac{\tilde{\epsilon}_{\mathbf{k}}}{mU}\right|^2 + \mathcal{O}\left(r^4\right)\right] \,, \end{aligned}$$

where we defined the ratio

$$r \equiv \frac{\tilde{t}^{(s^*)}}{mU} \,, \tag{1.112}$$

being $\tilde{\epsilon}_{\mathbf{k}} = \mathcal{O}\left(\tilde{t}^{(s^*)}\right)$.

By inserting this result in the self consistency equation for m, Eq. (1.68), we obtain

$$\begin{aligned} m &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \frac{mU}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, 0] \\ &\simeq \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \left[1 - \frac{1}{2}\left|\frac{\tilde{\epsilon}_{\mathbf{k}}}{mU}\right|^2 + \mathcal{O}\left(r^4\right)\right] \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \,. \end{aligned} \tag{1.113}$$

In last line we used Eq. (1.64), valid at half-filling. Let $m \neq 0$. From Eq. (1.111) we know that $\tilde{E}_{\mathbf{k}} \neq 0$. The hyperbolic tangent is, at low temperature, approximated by,

$$\lim_{\beta \to \infty} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) = 1 \,.$$

Therefore, Eq. (1.113) becomes

$$m \simeq \frac{1}{2} - \underbrace{\frac{1}{2 L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \left|\frac{\tilde{\epsilon}_{\mathbf{k}}}{mU}\right|^2}_{\mathcal{O}(r^2)} + \mathcal{O}\left(r^4\right) \,. \tag{1.114}$$

The second order correction can be made small in two ways:

1. By increasing U. From Eq. (1.112), we see that r decreases as U grows. This is the "conventional" way of magnetising the Hubbard lattice, by taking advantage of local repulsion which induces direct exchange (see App. **??**).

2. By shrinking bands. From Eq. (1.112), we see that as the renormalised bandwidth goes to zero, so does r. This is the "unconventional" way of magnetising the Hubbard lattice, where localisation is not necessarily due to a strong on-site repulsion, rather a low electronic mobility on the lattice.

Hence, as we wanted to prove, a strong Mott flattening can induce commensurate antiferromagnetism.

Approximation of free energy at half-filling. At the end of Sec. 1.2.3, we discussed the fact that the ground-state we find for the AF phase is stable only at half-filling, where $\tilde{\mu} = 0$. We can derive an approximation to free energy in this condition.

$$f^{(\mathrm{AF})} \simeq -\frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \tilde{E}_{\mathbf{k}} + Um^2 - V \sum_{\gamma} \left|u^{(\gamma)}\right|^2 . \tag{1.115}$$

This approximation is valid when $\tilde{E}_{\mathbf{k}} \neq 0$ everywhere. This is always realised at finite magnetisation, as is stated in Eq. (1.111). At half filling and low temperature, the system is well approximated by a completely filled electronic band $-\tilde{E}_{\mathbf{k}}$.

From a computational point of view, Eq. (1.115) can be useful when calculating the free energy of the ground-state, since the logarithms of Eq. (1.108) were removed.

Proof 1.22 — Proof of Eq. (1.115).

We start with the identity:

$$\begin{aligned}\ln\left(1 - \frac{1}{e^x + 1}\right) + \ln\left(1 - \frac{1}{e^{-x} + 1}\right) &= \ln\left[\frac{e^x}{(e^x + 1)^2}\right] \\ &= x - 2\ln(1 + e^x) .\end{aligned}$$

At $x \to \infty$, we have

$$x - 2\ln(1 + e^x) \simeq -x \qquad (x \to \infty) .$$

Let now $x = \beta\tilde{E}_{\mathbf{k}}$. At low temperature for a band with no nodes, we can approximate $\forall\mathbf{k} :\ \beta\tilde{E}_{\mathbf{k}} \to +\infty$ on the entire BZ. Hence

$$\ln\left(1 - \frac{1}{e^{\beta\tilde{E}_{\mathbf{k}}} + 1}\right) + \ln\left(1 - \frac{1}{e^{-\beta\tilde{E}_{\mathbf{k}}} + 1}\right) \simeq -\beta\tilde{E}_{\mathbf{k}} .$$

Substituting in Eq. (1.108), we obtain Eq. (1.115). □

1.6 Discussion of numeric results

In this section, we present numeric results. For the AF phase, the HFPs vector has two components:

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix} \quad \text{with} \quad \mathbf{v} = \mathbb{R}^2 .$$

The RMPs are, as for the normal phase:

$$\begin{aligned}\tilde{t}_{ij} &\equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)} , \\ \tilde{\mu} &\equiv \mu + n(2zV - U) .\end{aligned}$$

We simulate in the ranges:

$$0 \leq U \leq 8 , \qquad 0 \leq V \leq 8 ,$$

and limit to $\delta = 0$, since we restrict to the commensurate AF phase which is stable only at half-filling (see Sec. 1.2.3). Tolerances on density and HFPs have been set to

$$\Delta n = 10^{-2} , \qquad \Delta\mathbf{v} = \begin{bmatrix} m\colon 5\times 10^{-4} \\ u^{(s^*)}\colon 5\times 10^{-4} \end{bmatrix} .$$

In order to highlight the relation among all variables in the formation of isomagnetic domains, we show heatmaps and contour plots alongside, for a selection of six arbitrarily chosen contouring levels.

1.6.1 Mott localisation in the AF phase

In this subsection we present results for the HFP $u^{(s^*)}$ and the RMP $\tilde{t}^{(s^*)}$.

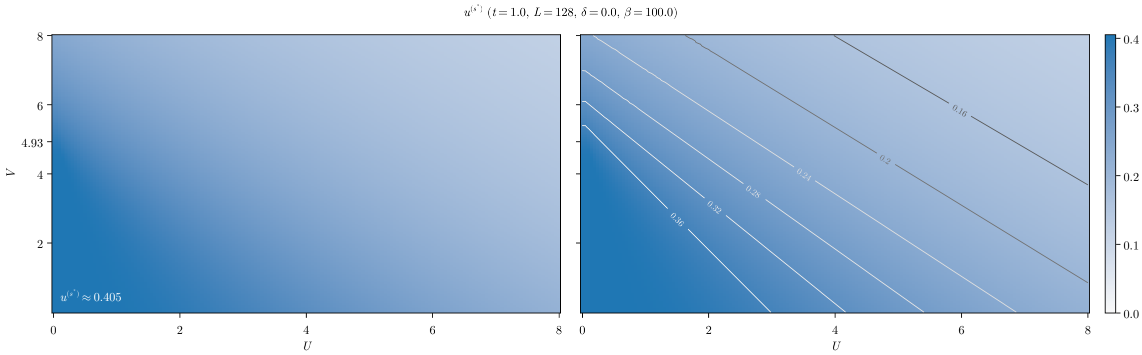
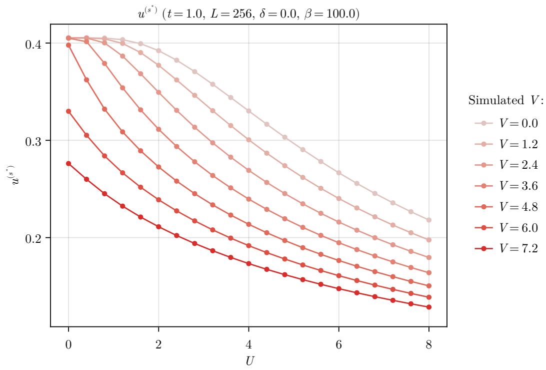
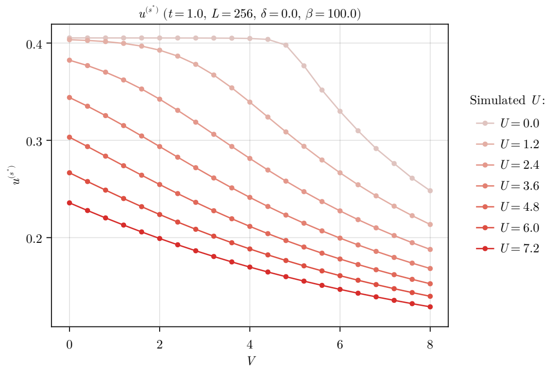


Figure 1.5 | Numerical results for the HFP $u^{(s^*)}$. In the left panel we indicate the converged value of $u^{(s^*)}$ in the bottom-left corner. In the right panel we plot isovalued lines.



a | U-dependency of $u^{(s^*)}$ varying V.



b | V-dependency of $u^{(s^*)}$ varying U.

Figure 1.6 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

Numerical results for $u^{(s^*)}$. In Fig. 1.5, we show the converged values of the HFP $u^{(s^*)}$ in the (U, V) plane at $t = 1.0$. We see in the left panel that $u^{(s^*)}$ is maximised in the bottom-left corner (low-U, low-V limit). As we see from the colour plot, $u^{(s^*)}$ exhibits a sort of triangle-shaped plateau in this region, where it reaches the value

$$u^{(s^*)} \lesssim 0.405\,.$$

The above converged value coincides with that computed at half-filling for the normal phase, in Eq. (**??**) . This is expected at $U = 0$, where magnetisation is necessarily null and we reduce to the normal phase. The approximate plateau in the bottom-left corner of Fig. 1.5 extends to $U/t \lesssim 2$, and indicates the region of parameters space where we expect magnetisation to be low enough to retrieve, approximately, the normal phase.

We indicated the tick $V = 4.93t$ on the vertical axis, where $U = 0$. We know from Eq. (**??**) that, at half-filling, it is the critical value for the Mott transition. Indeed, we see that the plateau of $u^{(s^*)}$ has the point

$$(U/t, V/t) = (0, 4.93)$$

as one of its "corners".

In the right panel of Fig. 1.5, we show a contour plot of the same data. We see that isovalued domains are, specifically, straight lines. Moreover, we see that as we move to the top-right corner (high-U, high-V) the isovalued lines become approximately parallel and $u^{(s^*)}$ decreases steadily.

The fact that $u^{(s^*)}$ is a decreasing function of both U and V is confirmed by looking at Fig. 1.6. In the left panel (Fig. 1.6a), we see that $u^{(s^*)}$ decreases as a function of U. For small V (brighter colour), the curves show a pleateau for $U \to 0$, which disappears upon increasing V. An analogous behaviour is observed in the right panel (Fig. 1.6b), by varying V at fixed values of U. We notice

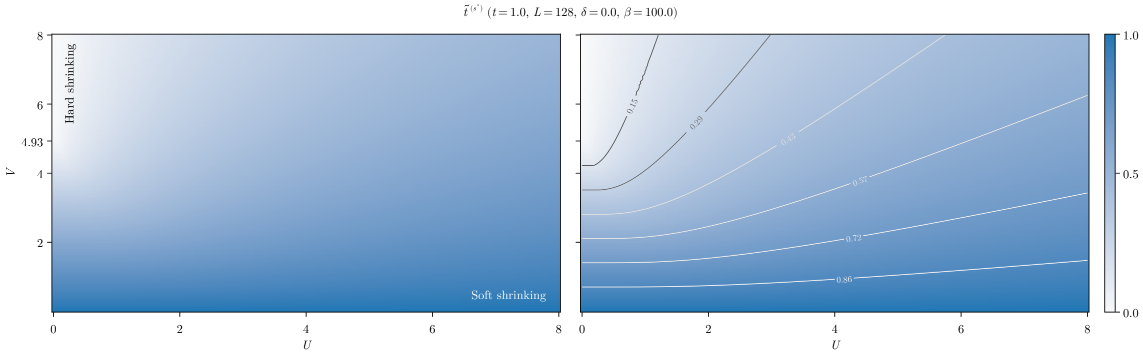
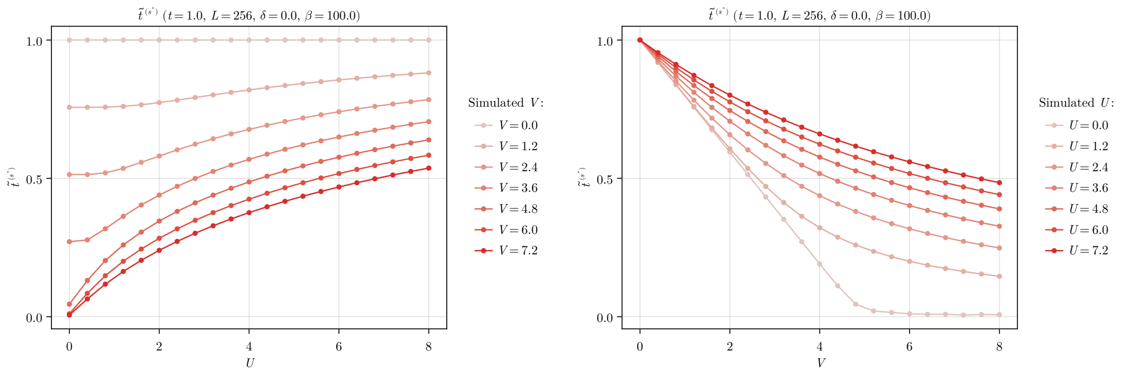


Figure 1.7 | Numerical results for the HFP $u^{(s^*)}$. In the left panel we indicate the converged value of $u^{(s^*)}$ in the bottom-left corner. In the right panel we plot isovalued lines.



a | U-dependency of $t^{(s^*)}$ varying V.

b | V-dependency of $t^{(s^*)}$ varying U.

Figure 1.8 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

that, since $m = 0$ for $U = 0$, the $U = 0$ curve of Fig. 1.6b coincides with the one at $\delta = 0$ for the normal phase, reported in Fig. **??**.

Hopping renormalisation. In Fig. 1.7, we show the the corresponding values for the renormalised hopping amplitude, defined as

$$\tilde{t}^{(s^*)} = t - \frac{u^{(s^*)}V}{2}\,.$$

As is clear from the definition, isovalued curves of $\tilde{t}^{(s^*)}$ and $u^{(s^*)}$ do not coincide. Specifically, we see that the top-left (low-U, high-V) and bottom-right (high-U, low-V) corners differ significantly (while, for $u^{(s^*)}$, they were similar enough). In the first case, $\tilde{t}^{(s^*)}$ is minimised; in the second, is maximised. The two corners correspond, respectively, to an "strong shrinking" and a "weak shrinking" of bare bands. The region of hard-shrinking, namely the brighter region of the panels of Fig. 1.7, extends from the critical value $(U/t, V/t) = (0, 4.93)$ to small values of the local repulsion, approximately $U \lesssim 1$. This is most visible in the contour plot on the right panel: while isovalued lines in the soft shrinking region are approximately parallel to the horizontal axis, moving towards the hard shrinking region they bend progressively.

Apart from the vertical axis, $U = 0$, the converged value of $\tilde{t}^{(s^*)}$ never vanishes (within numerical tolerance). This is better appreciated in Fig. 1.8, where we show data for $\tilde{t}^{(s^*)}$ as we vary U and V. In the right panel (Fig. 1.8b) we see that increasing V while U is fixed makes $\tilde{t}^{(s^*)}$ decrease monotonically, but at any $U \neq 0$ we do not reach in our simulation range a condition of total flattening. In the left panel (Fig. 1.8a), we see that as we increase U, the renormalised hopping amplitude gets larger.

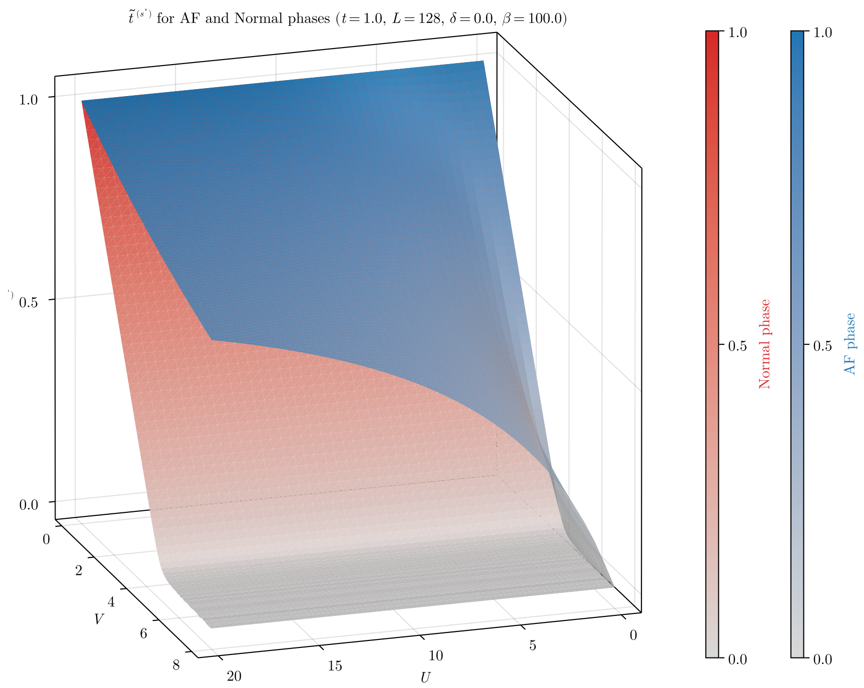


Figure 1.9 Comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase and the AF phase.

In Fig. 1.9 we show a comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase (red surface) and the AF phase (blue surface). Being the normal phase independent of U, the data for the normal phase are a concatenation at various values of the non-local repulsion of the $\delta = 0.0$ data shown in Fig. **??**. We see that the red surface reduces to zero only at $U = 0$, whereas the blue surface is consistently null at $V \gtrsim 4.93t$. As we mentioned before, in the AF phase we do not observe a complete flattening of bare bands. On the contrary, the presence of U reduces significantly the bands shrinking effect.

For the AF phase, we conclude that the non-local attraction $\hat{H}_V$ induces the flattening of bare bands, while the local repulsion $\hat{H}_U$ does the opposite. When we discussed Mott localisation in the context of the normal phase (see Sec. **??**), we argued that bands flattening was induced by V, while U was irrelevant. Here we see that in the AF phase, the local repulsion opposes Mott localisation. This is somewhat the opposite of what we would expect, based on the phenomenology of Mott localisation, which is in general enhanced by the repulsion of two electrons on the same site (see Sec. **??**).

The above observation need to be addressed in the context of antiferromagnetic ordering, where magnetisation m is a macroscopic observable. Recall the discussion at the end of Sec. 1.5: we claimed that the flattening of bare bands could induce magnetisation. Indeed, if by increasing U we are "by force" enhancing the AF gap (even at low magnetisation), by increasing V we are reducing the bare bandwidth. This suggests that we can obtain the same macroscopic magnetisation even if the underlying physics is much different. Indeed, as we observe from Fig. 1.8, it is only by increasing the non-local attraction that the actual hopping amplitude of electrons in reduced. From the above considerations, we observe that either increasing U or V we should expect approximately the same magnetic ordering.

1.6.2 Mott-induced enhancement of magnetisation

In Fig. 1.10 we show the converged results for m as a function of U and V. Looking at the left panel, we see that in the bottom-left corner (low-U, low-V) a bright triangle-shaped region is present in this area, where magnetisation is low. While we cannot assess precisely if this low-magnetisation region has the same shape of the plateau observed in the left panel of Fig. 1.5, we see that they

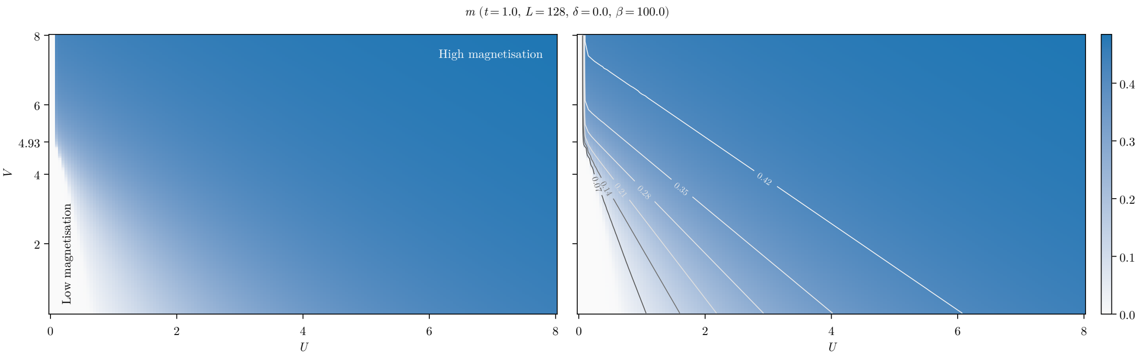
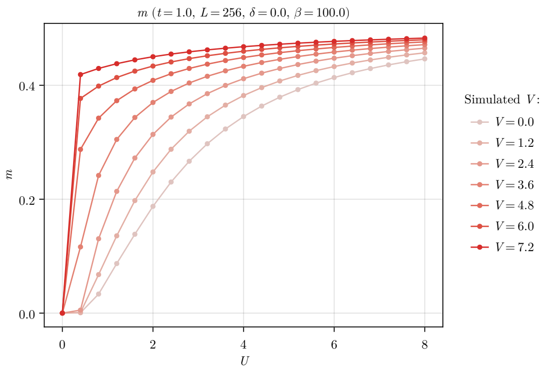
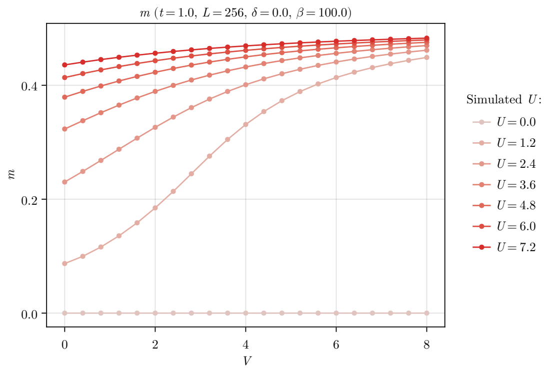


Figure 1.10 | Numerical results for the HFP $u^{(s^*)}$. In the left panel we indicate the converged value of $u^{(s^*)}$ in the bottom-left corner. In the right panel we plot isovalued lines.



a | U-dependency of m varying V.



b | V-dependency of m varying U.

Figure 1.11 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

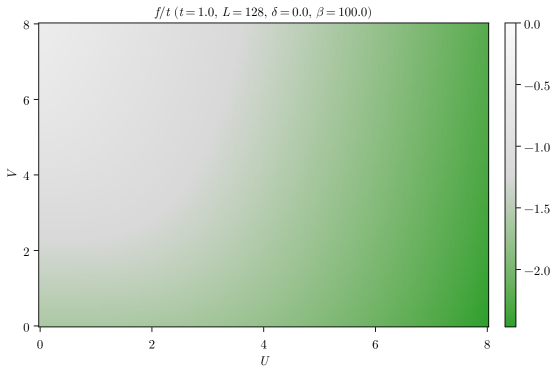
share the same top-left corner at $(U/t, V/t) = (0, 4.93)$. This is most evident in the contour plot on the right panel of Fig. 1.10. Indeed, consider a path starting from the low-m region and moving vertically, namely increasing V while keeping U fixed. If U is small enough, the transition of m to approximately zero to its saturation value happens as soon as we reach the hard-shrinking region highlighted in the left panel of Fig. 1.7.

Looking at the contour plot on the right panel of Fig. 1.10, we see that isomagnetic curves are actually straight lines, and become approximately parallel as we move towards the top-right corner of the plot. This is similar to what we found for the HFP $u^{(s^*)}$, and this suggests that isovalued curves for $u^{(s^*)}$ and m coincide. The fact that isovalued curves are linearly shaped suggests that possibly some kind of critical ratio among the interactions exists such that the net sum of local repulsion and Mott physics lead to the same macroscopic magnetic ordering. This relation should be, roughly, something like

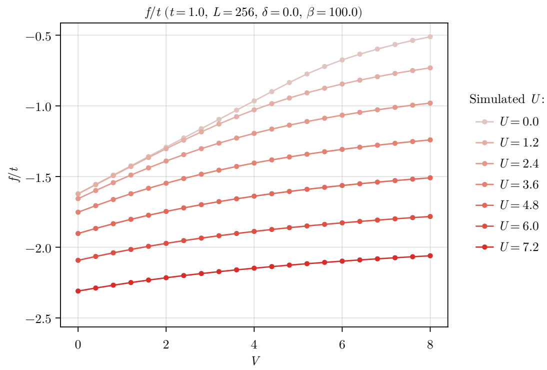
$$\frac{V}{V_0} = \frac{U_0 - U}{U_0},$$

where V_0 is the intersection of the isomagnetic line with the vertical axis ($U = 0$) and U_0 with the horizontal axis ($V = 0$).

In Fig. 1.11 we show the converged data for m as we vary, alternatively, U (left panel) and V (right panel). As we already commented, magnetisation is induced by both interactions, but the general shape of the curves is slightly different. In Fig. 1.11a we see that m saturates rapidly, but as we lower U all curves go to zero. In Fig. 1.11b, instead, lowering V the curves reach a finite intersection with the vertical axis. The local repulsion is necessary in order to establish antiferromagnetic ordering, while the non-local attraction acts – at fixed U – as a sort of magnetisation boost.



a | Heatmap of f in δV plane.



b | V-dependency of f varying δ.

Figure 1.12 | Free energy density of the AF phase.

1.6.3 Solution free energy density

In Fig. 1.12 the free energy density is plotted. In the right panel, Fig. 1.12b, we see that free energy is minimised in the bottom-right corner (low-V and high-U). In the top-left corner of Fig. 1.12a free energy grows, eventually reaching zero in the hard-shrinking region.

It is then evident how, even if the macroscopic magnetic properties are the same along one isomagnetic line of those plotted in Fig. 1.10, right panel, and thus the realised state exhibits an identical value of m, much different is the energetic content of the state itself. Increasing U while keeping V fixed, we amplify the gap. Hence, Bogoliubov bands $\pm\tilde{E}_{\mathbf{k}}$ are shifted away, namely the lower band is pushed towards more negative energies and the upper band towards more positive energies. At half-filling and low temperature, the lower band is roughly completely occupied while the upper band is roughly completely empty. Hence, quasiparticles lower their energy by the amplification of the gap. On the other hand, increasing V while keeping U fixed does the opposite. As bands get flattened, their distance lowers and quasiparticles increase their energies (namely, the lower band gets closer to zero). Hence, even if we observe isomagnetic domains, this does not mean that the content in energy of the HF ground state is identical.

The AF phase is always energetically favoured at half-filling, with respect to the normal phase. In Fig. 1.13 we plot the energy difference

$$f^{(\mathrm{N})} - f^{(\mathrm{AF})}\,.$$

The data for the normal phase, similarly to Fig. 1.9, are a concatenation of those of Fig. **??** at $\delta = 0.0$. We see that, on the entire $(U/t, V/t)$ region of our simulations, the free energy difference is positive. Hence the AF phase is consistently more stable with respect to the normal phase, except for the segment at $U = 0$, where the two coincide. Looking at Fig. 1.13, we see that the energy difference is enhanced as we increase V at fixed U.

At low temperature and half filling, it is not worth plotting entropy density. Half-filled commensurate antiferromagnets are insulators, with two gapped Bogoliubov bands. At low temperature, electronic distribution fills completely the lower band and empties the upper one. Because of this, insulating gapped systems inherently are low entropic and our results are compatible with zero in the entire region of parametric space we explored.

1.6.4 Fixed-point interpretation of HF solution

We end this chapter, by presenting a final remark regarding the numerical structure of the algorithm. As we anticipated in Sec. **??**, our convergence scheme can be seen as an effective search for the singular fixed point of a nonlinear map. A physical solution is represented as a HFP vector $\mathbf{v}$ such that the algorithm is a local identity,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v}\,.$$

To predict for a given initialiser $\mathbf{v}_0$ the path it will follow through HFPs space is, in general, highly non-trivial due to the intrinsic nonlinearity of the algorithmic map defined by self-consistency

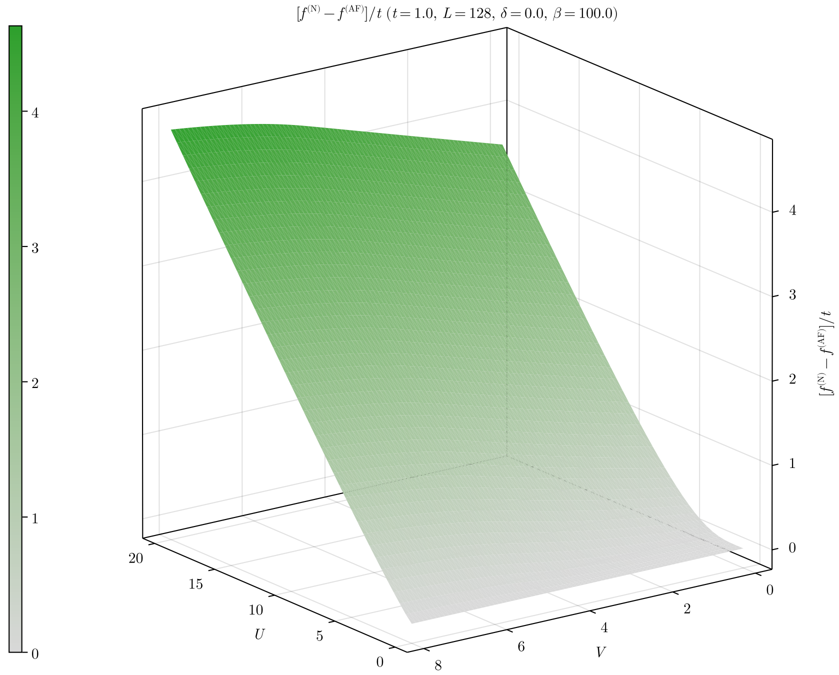


Figure 1.13 | Difference of free energies for the normal phase minus the AF phase.

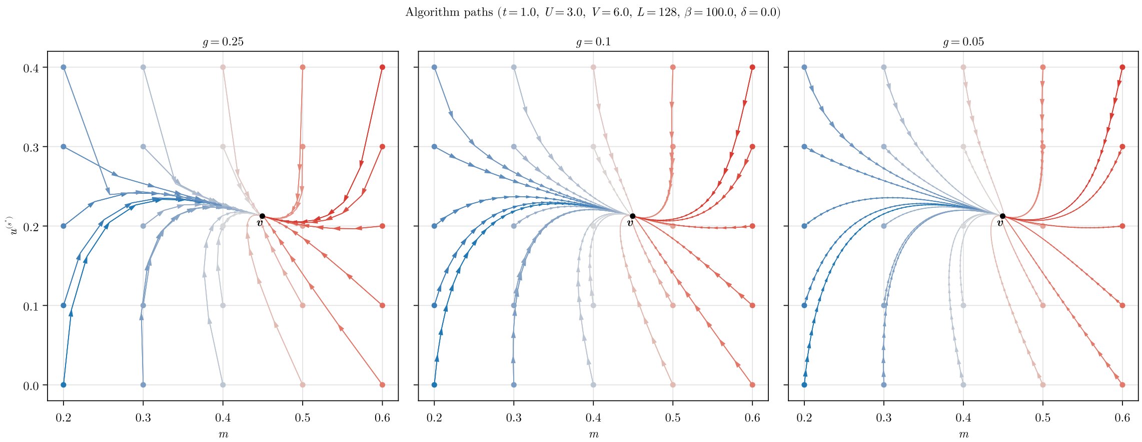


Figure 1.14 | Quiver plot of algorithmic evolution in HFPs space at fixed model parametrisation. The black dot **v** represents the nonlinear map fixed point, while the colored paths the sequences of HFPs coordinates simulated via iterative HF procedure, starting from different HFPs initialisers (colored dots).

equations for the various HFPs. An example of such nonlinearity is given in the derivation of Secs. **??**, **??**.

In Fig. 1.14 we sketch this concept in a practical situation. In figure are plotted various algorithmic paths through AF HFPs space, with operational setup at

$$U = 3\,, \qquad V = 6\,, \qquad L = 128\,, \qquad \beta = 100\,, \qquad \delta = 0\,.$$

At this point, the self-consistent HF “fixed-point” solution is given by

$$m \approx 0.449\,, \qquad u^{(s^*)} \approx 0.212\,,$$

indicated in panels as the black dot. The three panels show different map evolutions varying the mixing parameter within three choices, $g \in \{0.25, 0.1, 0.05\}$. We chose various initialisers in the set:

$$\mathbf{v}_0 = \begin{bmatrix} m_0 \\ u_0^{(s^*)} \end{bmatrix} \quad : \quad m_0 \in \{0.2,\ 0.3,\ ...,\ 0.6\} \quad \text{and} \quad u_0^{(s^*)} \in \{0.0,\ 0.1,\ ...,\ 0.4\}\,.$$

In figures, different colours indicate different evolutions, while the tip size indicates the magnitude of the algorithmic step in passing from one iteration to the following.

All paths are able to converge to $\mathbf{v}$, and the quantity of steps necessary to reach the fixed point, a sort of attractor, increases as we lower g. To select the correct g turns out to be the real algorithm fine-tuning knob: as we saw for the normal phase, some model parametrisation could lead to unavoidable algorithmic instabilities. The problem is that we do not know beforehand where the instabilities will occur, and how. The correct g is the compromise between keeping low the number of steps (larger mixing) and moving slow enough to not diverge when encountering unstable points (smaller mixing). By recursively fine-tuning g repeating non-converged simulation with more care, we are essentially making the nonlinear map "softer" and curvier, as is in the right panel of Fig. 1.14. It's not always resource convenient to do so, so for most simulations we stick to a rougher map (like the left panel one) by setting $g = 0.5$.

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